QUALIFICATION REQUIREMENTS REMOVED

MIL-M-38510/460A(USAF)
29 June 1984
SUPERSEDING
MIL-M-38510/460(USAF)
30 March 1979

MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, INTEGRATED INJECTION LOGIC (I^2L), PARALLEL 16-BIT MICROPROCESSOR, MONOLITHIC SILICON

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS REVISION.

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for a monolithic silicon, integrated injection logic (I^2L), parallel 16-bit microprocessor. One product assurance class and a choice of lead finish are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510, with the exception that the "JAN" or "J" certification shall not be used.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	16-bit fixed instruction microprocessor

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
Y	See figure 1 (64-lead, 0.8" x 3.2"), dual-in-line package

1.3 Absolute maximum ratings:

Supply current, I_{CC}	750 mA
Input voltage, V_{IN} (all except INJ input)	+5.5 V dc
V_{INJ}	+2.0 V dc
Power dissipation	750 mW
Junction temperature (T_J)	+175°C
Thermal resistance, junction to case (θ_{JC})	25°C/W
Lead temperature (soldering, 10 seconds)	+300°C
Storage temperature range	-65°C to +150°C

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions: (Unless otherwise noted, I_{CC} = 500 mA.)

Supply current (I_{CC})	500 mA minimum to 550 mA maximum
High-level input voltage (V_{IH})	2.0 V minimum
Low-level input voltage (V_{IL})	0.8 V maximum
Frequency of operation	DC to 3 MHz maximum
Case operating temperature range	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specifications. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Terminal connections. Terminal connections (pin assignments) shall be as specified on figure 2.

3.2.2 Block diagram. The block diagram (architecture) shall be as specified on figure 3; the microprocessor flow chart is shown on figure 4.

3.2.3 Memory map. The memory map shall be as specified on figure 5.

3.2.4 General features, functional description, and instruction set. The general features, functional description, and instruction set (with terms and symbols) shall be as specified in the appendix to this specification.

3.2.5 Case outline. The case outline shall be in accordance with 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test		Symbol	Conditions	Limits Min	Limits Max	Unit
High-level input voltage		V_{IH}		2		V
Low-level input voltage		V_{IL}			0.8	V
Input clamp voltage		V_{IC}	I_{CC} = 500 mA, I_{IN} = 12 mA		-1.5	V
High-level output current	I/O pins	I_{OH5}	I_{CC} = 500 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V, V_{OH} = 2.4 V		300	µA
	Other outputs	I_{OH4} I_{OH6}			50	µA
	I/O pins	I_{OH2}	I_{CC} = 500 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V, V_{OH} = 5.5 V		1	mA
	Other outputs	I_{OH1} I_{OH3}			250	µA
Low-level output voltage		V_{OL}	I_{CC} = 500 mA, V_{IH} = 2.0 V, V_{IL} = 0.8 V, I_{OL} = 20 mA		0.4	V
Input current	Clock	I_{IH}	I_{CC} = 500 mA, V_{IN} = 2.4 V		1,000	µA
	Other inputs				300	µA
Setup time to rising edge of clock	READY	t_{SHL2}	See figure 11	70		ns
	DO-D15	t_{SHL3}		65		
	CRUIN	t_{SHL4}		50		
	$\overline{INTREQ}$	t_{SHL5}		25		
	IC0-IC3	t_{SHL6}		25		
Hold time from rising edge of clock	READY	t_{HHL2}	See figure 11	50		ns
	DO-D15	t_{HHL3}		40		
	CRUIN	t_{HHL4}		40		
	$\overline{INTREQ}$	t_{HHL5}		55		
	IC0-IC3	t_{HHL6}		55		

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TABLE I. Electrical performance characteristics - Continued.

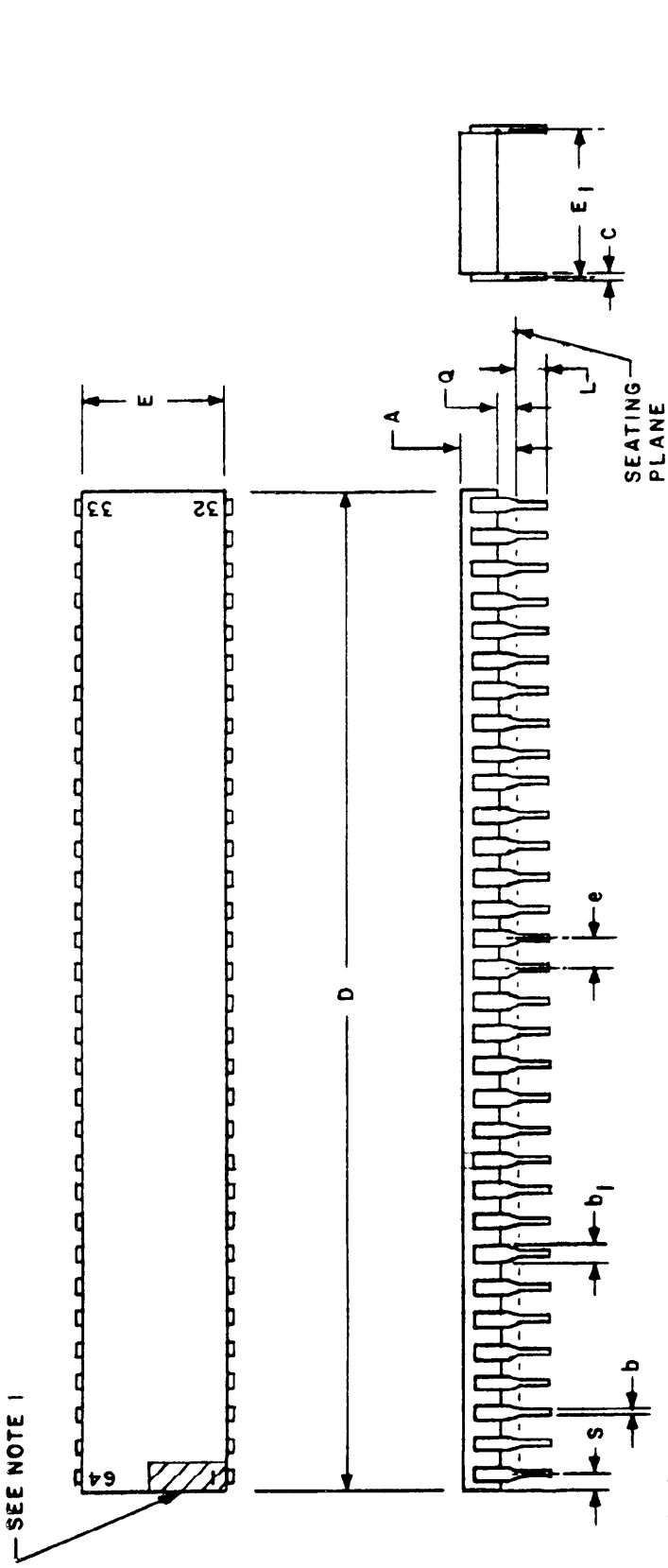
Test	Symbol	Conditions	Limits		Unit
			Min	Max	
Frequency of operation	F1		3		MHz
Minimum clock high time	t_{PWH}		222		ns
Minimum clock low time	t_{PWL}		111		ns
Delay times from clock to:					
Address bus (A0-A14)	t_{PHL1}	CL = 150 pF ±5% See figures 8 and 10		140	ns
Data bus (D0-D15)	t_{PHL2}			155	
Write enable ($\overline{WE}$)	t_{PHL3}			185	
Cycle end ($\overline{CYCEND}$)	t_{PHL4}			145	
Data bus in (DBIN)	t_{PHL5}			160	
Memory enable ($\overline{MEMEN}$)	t_{PHL6}			140	
CRU clock (CRUCLK)	t_{PHL7}			155	
CRU data out (CRUOUT)	t_{PHL8}			185	
Hold acknowledge (HLDA)	t_{PHL9}			290	
Wait	t_{PHL10}			130	
Instruction acquisition (IAQ)	t_{PHL11}			130	

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)
	Class B devices
Interim electrical parameters (pre burn-in) (method 5004)	1, 7, 9
Final electrical test parameters 1/ (method 5004)	1, 2, 3, 7, 8, 9, 10, 11
Group A test requirements (method 5005)	1, 2, 3, 7, 8, 9, 10, 11
Groups C and D end point electrical parameters (method 5005)	1, 7, 9
Additional electrical subgroups for group C periodic inspections	None

1/ PDA applies to subgroups 1, 7, and 9 (see 4.3f.).

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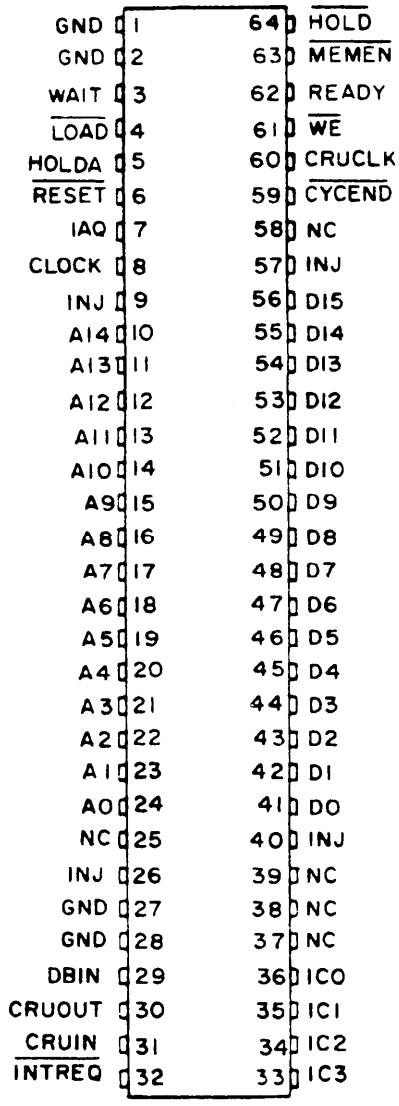


DIM	INCHES		MM		Notes
	Min	Max	Min	Max	
A		.225		5.72	
b	0.015	0.020	0.38	0.51	3
b_1		0.070		1.77	2,8
c	.008	.013	0.20	0.33	8
D	3.168	3.232	80.47	82.10	4
E	.880	.900	22.35	22.86	4
E_1	.890	.910	22.61	23.11	7
e	0.100	BSC	2.54	BSC	5,9
L	0.100	0.165	2.54	4.19	
Q	0.020	0.060	0.51	1.52	3
S	.020	.065	0.51	1.65	6
α	0°	15°			

NOTES:
1. Index area; a notch or a pin one identification mark shall be located adjacent to pin one and shall be located within the shaded area shown. The manufacturer's identification shall not be used as a pin one identification mark.
2. The minimum limit for dimension b_1 may be .020 (.51 mm) for leads number 1, 32, 33 and 64 only.
3. Dimension Q shall be measured from the seating plane to the base plane.
4. This dimension allows for off-center lid, meniscus and glass overrun.
5. The basic pin spacing is .100 (2.54 mm) between centerlines. Each pin centerline shall be located within ±.010 (.25 mm) of its exact longitudinal position relative to pin 1 and 64.
6. Applies to all four corners (leads number 1, 32, 33 and 64).
7. E_1 shall be measured at the centerline of the leads.
8. All leads - increase maximum limit by .003 (.08 mm) measured at the center of the flat, when lead finish A is applied.
9. Sixty two spaces.

FIGURE 1. Case outline Y (64-lead, 0.8 "X 3.2").

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NC-NO INTERNAL CONNECTION

FIGURE 2. Terminal connections (pin assignments).

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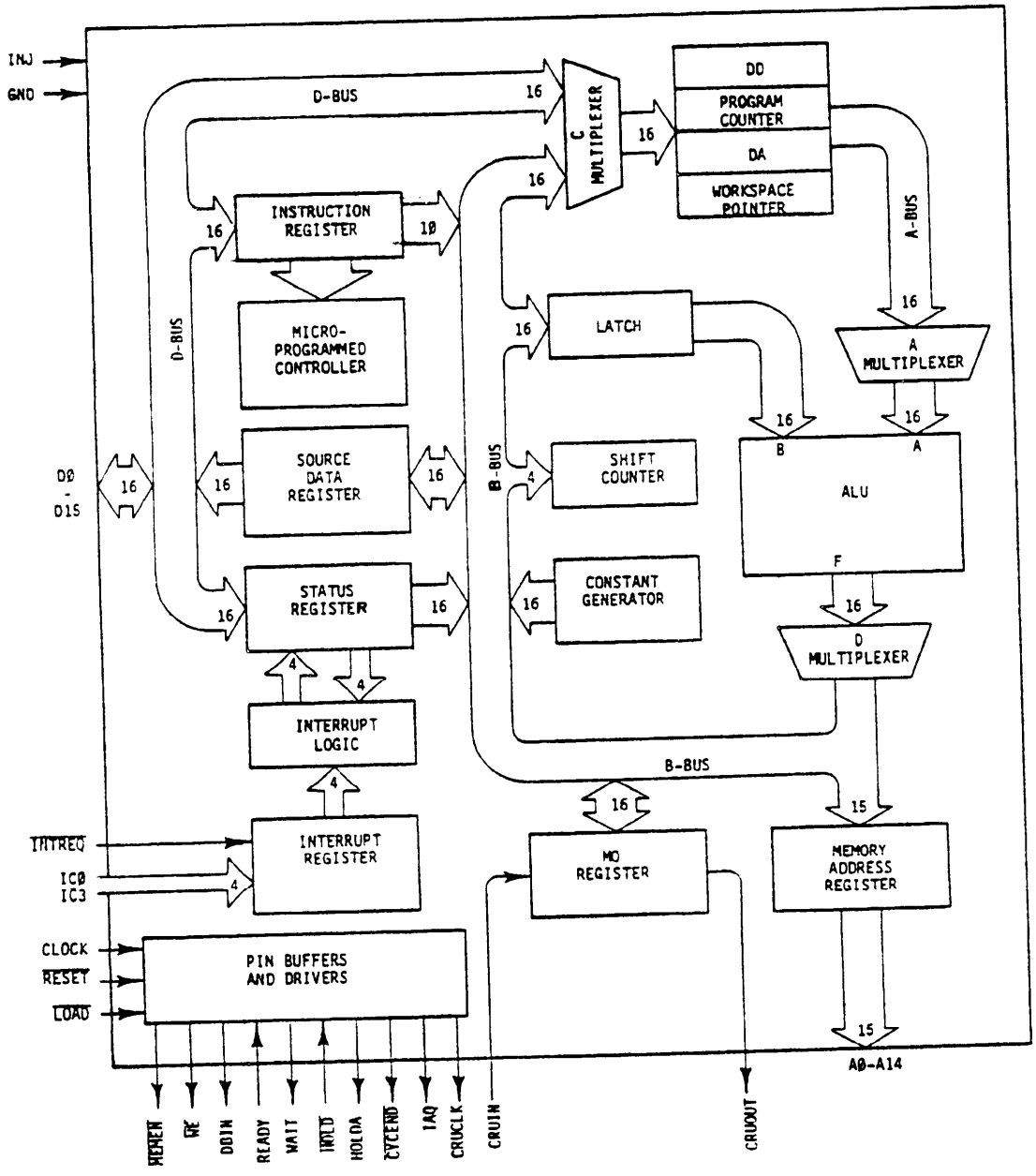


FIGURE 3. Block diagram (architecture).

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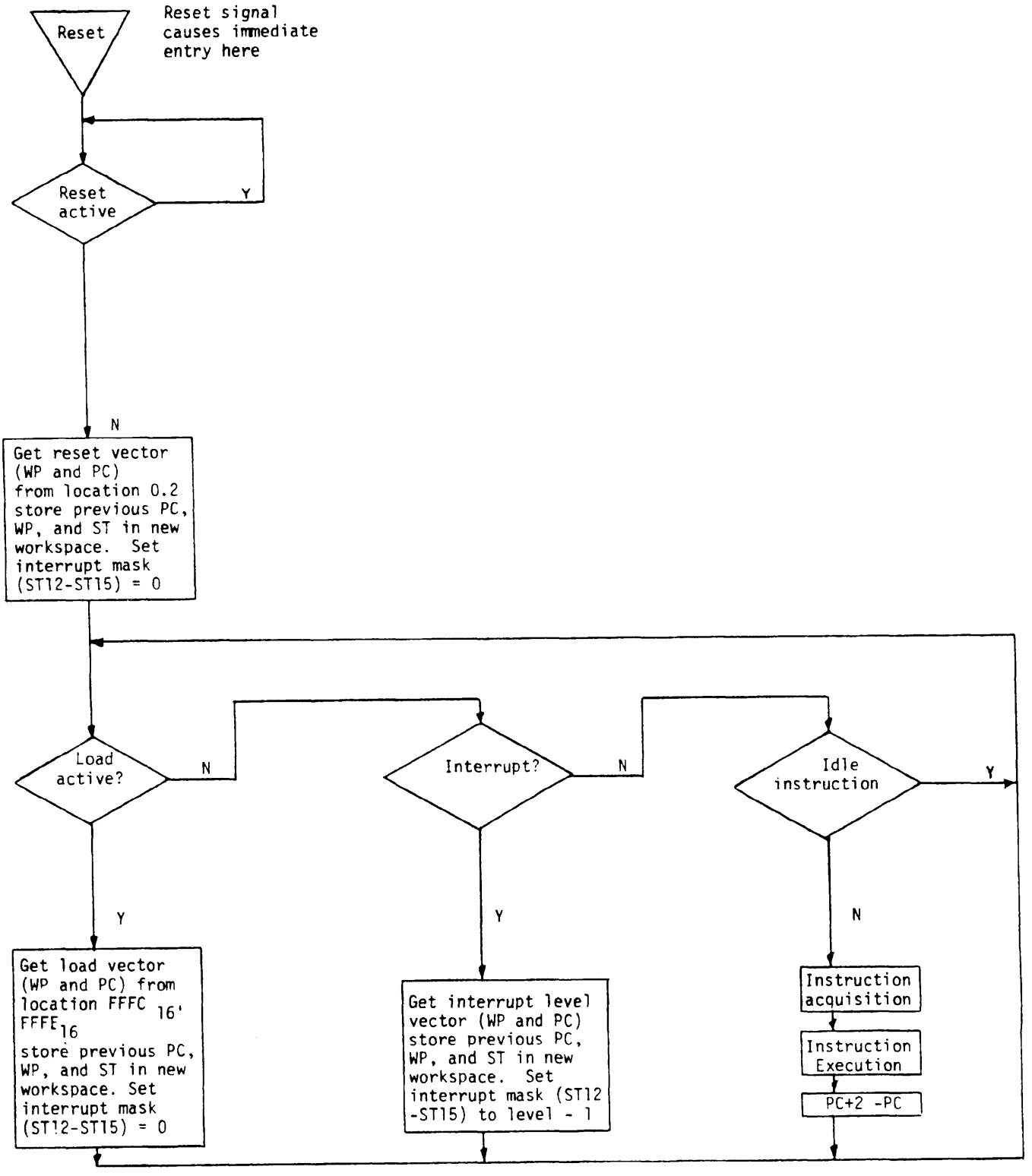


FIGURE 4. Microprocessor flow chart.

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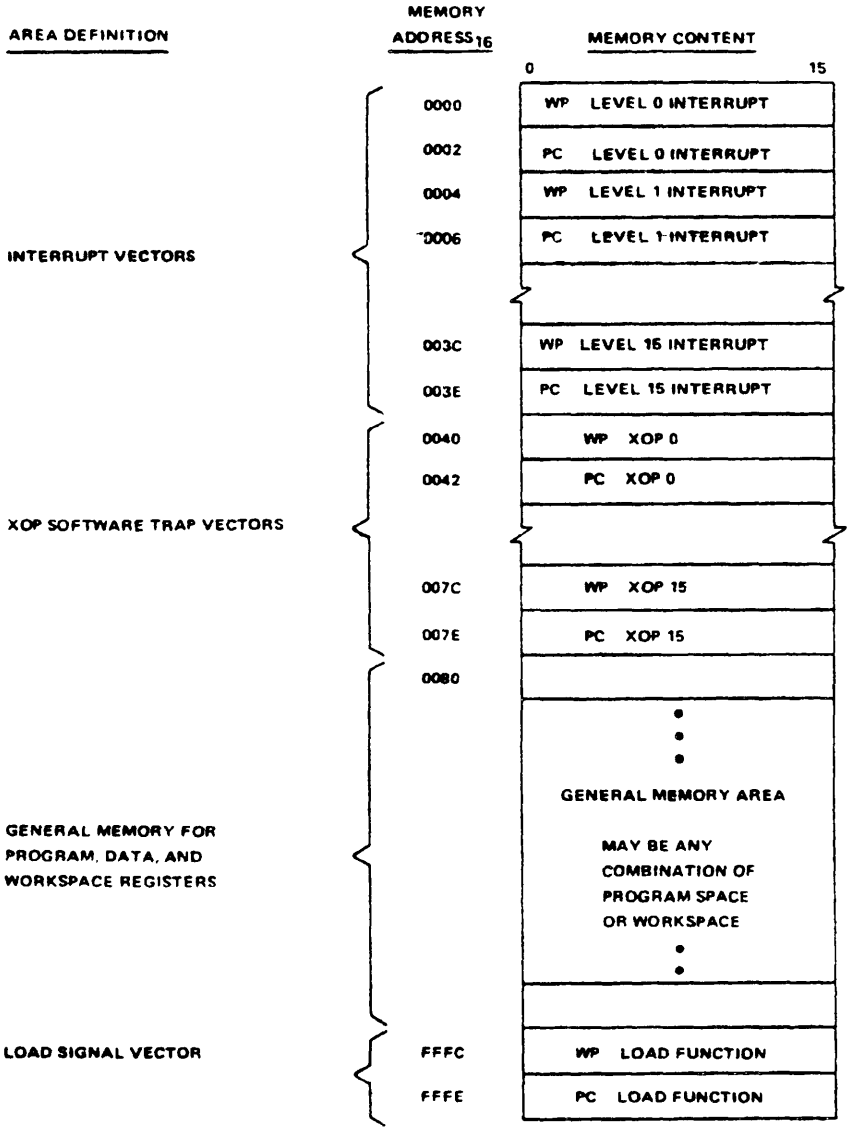
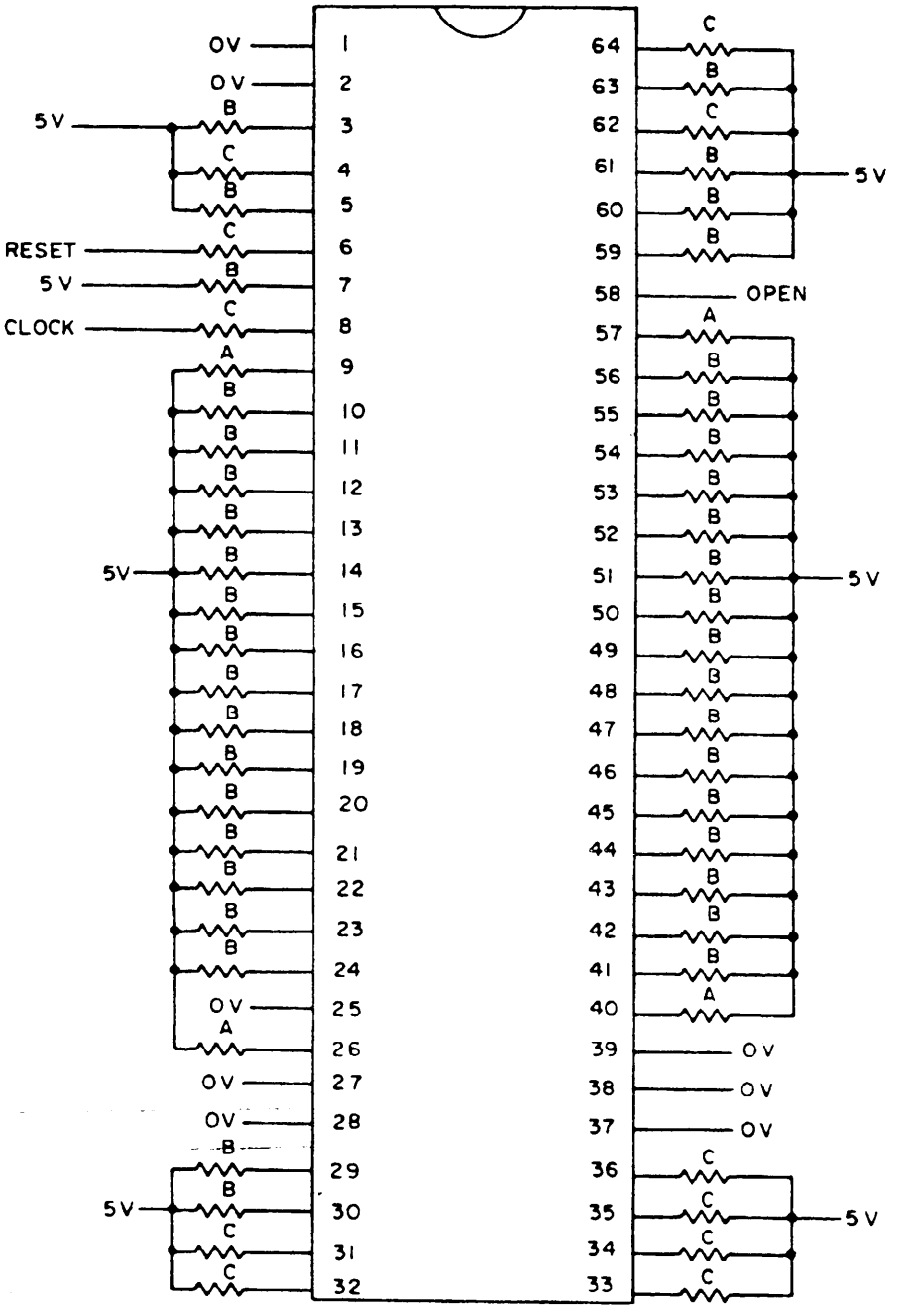


FIGURE 5. Memory map.

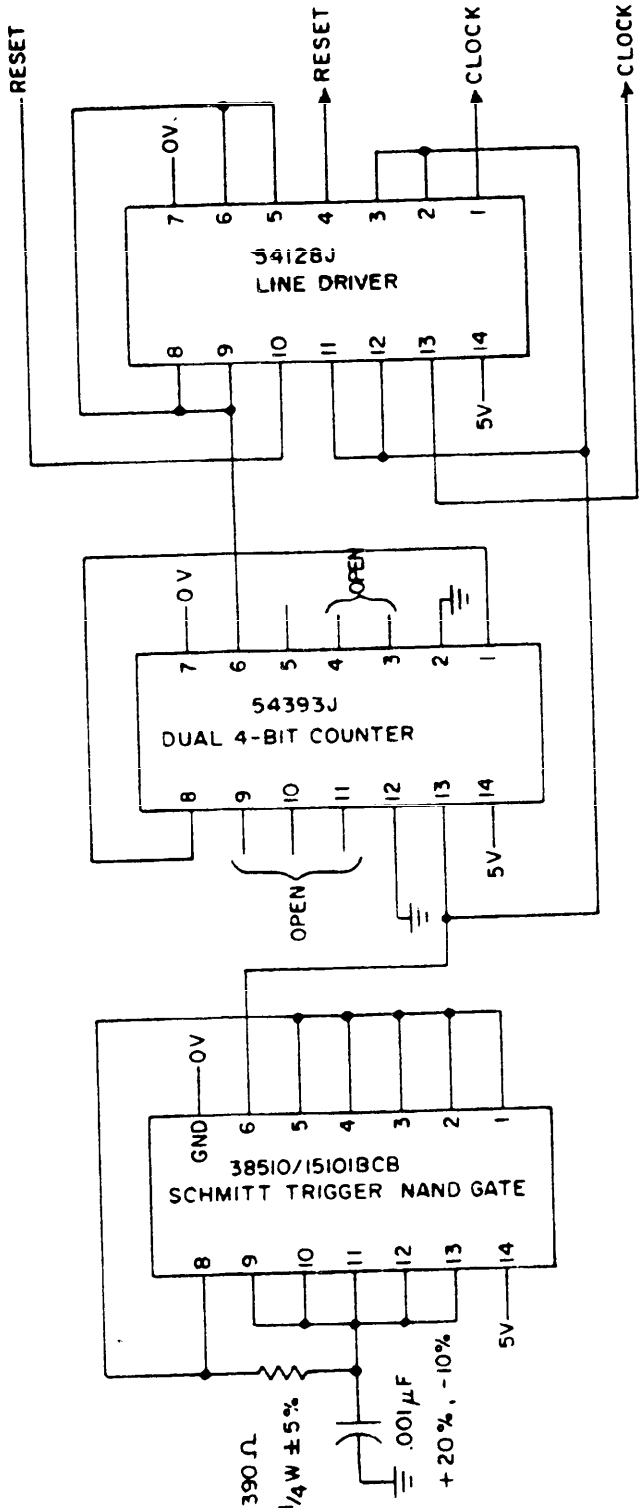
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NOTES:
1. All resistor values are in ohms:
   A = 33Ω, B = 330Ω, C = 1 kΩ.
2. Pins 9, 26, 40 and 57 are injector sources.
3. The four 33-ohm resistors are metal film 1 watt ±5% tolerance, and all other resistors are ≥ 1/4 watt ±5%.
4. See figure 7 for reset and clock circuit.

FIGURE 6. Burn-in and operating life test circuit.

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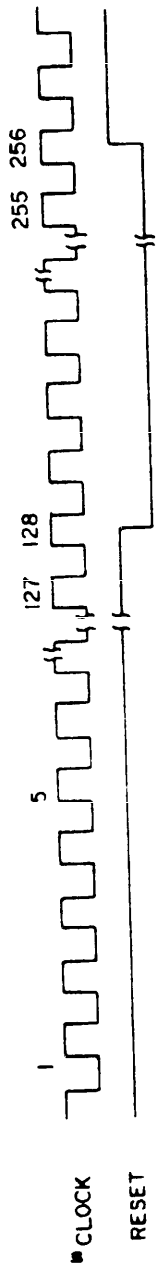


FIGURE 7. Reset/clock circuit for burn-in and operating life test.

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TABLE III. Group A inspection.
Terminal conditions (pins not designated are open) 1/

Subgroup	Symbol	MIL-STD-883 method	Test No.	1, 2, 27, 28	9, 26, 40, 57	3	4	5	6	7	8	10-24	29	30	31	32	33-36	41-56	59	60	61	62	63	64	Measured terminal	Test limits		Unit
				GND	INJ	WAIT	LOAD	HOLDA	RESET	IAO	CLOCK	A14-A0	DBIN	CRUDUT	CRUIN	INTREQ	IC3-IC0	DO-DI5	CYCLERQ	CRUCLK	RE	READY	MEMWR	HOLD		Min	Max	
1 T_C = +25°C	V_{IC}		1	GND	500 mA		-12 mA		-12 mA		12 mA				-12 mA										4		1.5	V
			2	"	"		"		"		"				"										6		"	"
			3	"	"		"		"		"				"										8		"	"
			4	"	"		"		"		"				"										31		"	"
			5	"	"		"		"		"				"	-12 mA									32		"	"
			6	"	"		"		"		"				"	"									33		"	"
			7	"	"		"		"		"				"	"									34		"	"
			8	"	"		"		"		"				"	"									35		"	"
			9	"	"		"		"		"				"	"									36		"	"
			10	"	"		2.0 V		A		A35				2.0 V	"	-12 mA					2.0 V		2.0 V	41		"	"
			11	"	"		"		"		"				"	"	"					"		"	42		"	"
			12	"	"		"		"		"				"	"	"					"		"	43		"	"
			13	"	"		"		"		"				"	"	"					"		"	44		"	"
			14	"	"		"		"		"				"	"	"					"		"	45		"	"
			15	"	"		"		"		"				"	"	"					"		"	46		"	"
			16	"	"		"		"		"				"	"	"					"		"	47		"	"
			17	"	"		"		"		"				"	"	"					"		"	48		"	"
			18	"	"		"		"		"				"	"	"					"		"	49		"	"
			19	"	"		"		"		"				"	"	"					"		"	50		"	"
			20	"	"		"		"		"				"	"	"					"		"	51		"	"
			21	"	"		"		"		"				"	"	"					"		"	52		"	"
			22	"	"		"		"		"				"	"	"					"		"	53		"	"
			23	"	"		"		"		"				"	"	"					"		"	54		"	"
			24	"	"		"		"		"				"	"	"					"		"	55		"	"
			25	"	"		"		"		"				"	"	"					"		"	55		"	"
			26	"	"		"		"		"				"	"	"					-12 mA		"	62		"	"
			27	"	"		"		"		"				"	"	"					"		-12 mA	64		"	"
	I_{OH1}		28	"	"	5.5 V	2.0 V		B		877					2.0 V		B				0.8 V		2.0 V	3		250	μA
			29	"	"		"	5.5 V	"		892					"		"				2.0 V		0.8 V	5		"	"
			30	"	"		"		"		877					"		"				0.8 V		2.0 V	7		"	"
			31	"	"		"		"		892	5.5 V				"		"				2.0 V		0.8 V	10		"	"
			32	"	"		"		"		"	"				"		"				"		"	11		"	"
			33	"	"		"		"		"	"				"		"				"		"	12		"	"
			34	"	"		"		"		"	"				"		"				"		"	13		"	"
			35	"	"		"		"		"	"				"		"				"		"	14		"	"
			36	"	"		"		"		"	"				"		"				"		"	15		"	"
			37	"	"		"		"		"	"				"		"				"		"	16		"	"
			38	"	"		"		"		"	"				"		"				"		"	17		"	"
			39	"	"		"		"		"	"				"		"				"		"	18		"	"
			40	"	"		"		"		"	"				"		"				"		"	19		"	"
			41	"	"		"		"		"	"				"		"				"		"	20		"	"
			42	"	"		"		"		"	"				"		"				"		"	21		"	"
			43	"	"		"		"		"	"				"		"				"		"	22		"	"
			44	"	"		"		"		"	"				"		"				"		"	23		"	"
			45	"	"		"		"		"	"				"		"				"		"	24		"	"
			46	"	"		"		"		"	"	5.5 V			0.8 V		"				"		"	29		"	"
			47	"	"		"		"		875	"		5.5 V		"		B				"		2.0 V	30		"	"
	I_{OH2}		48	"	"		"		"		882					"		"				"		0.8 V	41		1	mA
			49	"	"		"		"		"					"		"				"		"	42		"	"
			50	"	"		"		"		"					"		"				"		"	43		"	"
			51	"	"		"		"		"					"		"				"		"	44		"	"
			52	"	"		"		"		"					"		"				"		"	45		"	"
			53	"	"		"		"		"					"		"				"		"	46		"	"
			54	"	"		"		"		"					"		"				"		"	47		"	"
			55	"	"		"		"		"					"		"				"		"	48		"	"

See footnotes at end of table.

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TABLE III. Group A tests. Continued.
Terminals conditions (pins not designated are open) 1/

Subgroup	Symbol	MIL-STD-883 method	Test No.	1, 2, 27, 28	9, 26, 40, 57	3	4	5	6	7	8	10-24	29	30	31	32	33-36	41-56	59	60	61	62	63	64	Measured terminal	Test limits Min	Test limits Max	Unit
				GND	INJ	WAIT	LOAD	HOLDA	RESET	IAQ	CLOCK	A14-A0	DBIN	CRJOUT	CRUIN	INTREQ	IC3-IC0	DO-D15	CYCENO	CRUCLK	WE	READY	MEMEN	WDCO				
1 T_C = +25°C	I_{OH2}		56	GND	500 mA		2.0 V		B		BR2					2.0 V		B				2.0 V		0.8 V	49		1	mA
			57	"	"		"		"		"					"		"				"		"	50		"	"
			58	"	"		"		"		"					"		"				"		"	51		"	"
			59	"	"		"		"		"					"		"				"		"	52		"	"
			60	"	"		"		"		"					"		"				"		"	53		"	"
			61	"	"		"		"		"					"		"				"		"	54		"	"
			62	"	"		"		"		"					"		"				"		"	55		"	"
			63	"	"		"		"		"					"		"				"		"	56		"	"
	I_{OH3}		64	"	"		"		"							"		"	5.5 V			"		"	59		250	μA
			65	"	"		"		"		B75					"		"		5.5 V		"		2.0 V	60		"	"
			66	"	"		"		B		BR2					"		"			5.5 V	"		0.8 V	61		"	"
			67	"	"		2.0 V		"		BR2					2.0 V		"				2.0 V	5.5 V	0.8 V	63		"	"
	I_{OH4}		68	"	"	2.4 V	"		"		B77					"		"				0.8 V		2.0 V	3		50	"
			69	"	"		"	2.4 V	"		BR2					"		"				2.0 V		0.8 V	5		"	"
			70	"	"		"		"	2.4 V	B77					"		"				0.8 V		2.0 V	7		"	"
			71	"	"		"		"		BR2	2.4 V				"		"				2.0 V		0.8 V	10		"	"
			72	"	"		"		"		"	"				"		"				"		"	11		"	"
			73	"	"		"		"		"	"				"		"				"		"	12		"	"
			74	"	"		"		"		"	"				"		"				"		"	13		"	"
			75	"	"		"		"		"	"				"		"				"		"	14		"	"
			76	"	"		"		"		"	"				"		"				"		"	15		"	"
			77	"	"		"		"		"	"				"		"				"		"	16		"	"
			78	"	"		"		"		"	"				"		"				"		"	17		"	"
			79	"	"		"		"		"	"				"		"				"		"	18		"	"
			80	"	"		"		"		"	"				"		"				"		"	19		"	"
			81	"	"		"		"		"	"				"		"				"		"	20		"	"
			82	"	"		"		"		"	"				"		"				"		"	21		"	"
			83	"	"		"		"		"	"				"		"				"		"	22		"	"
			84	"	"		"		A		"	"				"		"				"		"	23		"	"
			85	"	"		"		"		"	"				"		"				"		"	24		"	"
			86	"	"		"		"		"		2.4 V			"	0.8 V	"				"		"	29		"	"
			87	"	"		"		B		B75			2.4 V		"		"				"		2.0 V	30		"	"
	I_{OH5}		88	"	"		"		A		BR2					"		"				"		0.8 V	41		100	"
			89	"	"		"		"		"					"		"				"		"	42		"	"
			90	"	"		"		"		"					"		"				"		"	43		"	"
			91	"	"		"		"		"					"		"				"		"	44		"	"
			92	"	"		"		"		"					"		"				"		"	45		"	"
			93	"	"		"		"		"					"		"				"		"	46		"	"
			94	"	"		"		"		"					"		"				"		"	47		"	"
			95	"	"		"		"		"					"		"				"		"	48		"	"
			96	"	"		"		"		"					"		"				"		"	49		"	"
			97	"	"		"		"		"					"		"				"		"	50		"	"
			98	"	"		"		"		"					"		"				"		"	51		"	"
			99	"	"		"		"		"					"		"				"		"	52		"	"
			100	"	"		"		"		"					"		"				"		"	53		"	"
			101	"	"		"		"		"					"		"				"		"	54		"	"
			102	"	"		"		"		"					"		"				"		"	55		"	"
			103	"	"		"		"		"					"		"				"		"	56		"	"
	I_{OH6}		104	"	"		"		"		"					"		"	2.4 V			"		"	59		50	"
			105	"	"		"		B		B75					"		"		2.4 V		"		2.0 V	60		"	"
			106	"	"		"		0.8 V		B82					"		"			2.4 V			0.8 V	61		"	"
			107	"	"		2.0 V		A		BR2					2.0 V		"				2.0 V	2.4 V	0.8 V	63		"	"

See footnotes at end of table.

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TABLE III. Group A Inspection - Continued.
Terminals conditions (pins not designated are open) 1/

Subgroup	Symbol	MIL-STD-883 method	Test No.	1, 2, 27, 28	9, 26, 40, 57	3	4	5	6	7	8	10-24	29	30	31	32	33-36	41-56	59	60	61	62	63	64	Measured terminal	Test limits		Unit
				GND	INJ	WAIT	LOAD	HOLDA	RESET	IAQ	CLOCK	A18-A0	DBIN	CRUOUT	CRUIN	INTREQ	IC3-IC0	DO-D15	CYCENO	CRUCLK	WE	READY	MEMEN	HOLD		Min	Max	
1 T_C = +25°C	V_{OL}	3007	108	GND	500 mA	20 mA	2.0 V	20 mA	A	20 mA	B35					2.0 V		B				2.0 V		2.0 V	3		400	mV
			109	"	"		"		"		"					"		"				"		"	5		"	
			110	"	"		"		"		"					"		"				"		"	7		"	
			111	"	"		"		"		B7	20 mA				"		"				"		"	10		"	
			112	"	"		"		B		"	"				"		"				"		"	11		"	
			113	"	"		"		"		"	"				"		"				"		"	12		"	
			114	"	"		"		"		"	"				"		"				"		"	13		"	
			115	"	"		"		"		"	"				"		"				"		"	14		"	
			116	"	"		"		"		"	"				"		"				"		"	15		"	
			117	"	"		"		"		"	"				"		"				"		"	16		"	
			118	"	"		"		"		"	"				"		"				"		"	17		"	
			119	"	"		"		"		"	"				"		"				"		"	18		"	
			120	"	"		"		"		"	"				"		"				"		"	19		"	
			121	"	"		"		"		"	"				"		"				"		"	20		"	
			122	"	"		"		"		"	"				"		"				"		"	21		"	
			123	"	"		"		"		"	"				"		"				"		"	22		"	
			124	"	"		"		"		"	"				"		"				"		"	23		"	
			125	"	"		"		"		"	"				"		"				"		"	24		"	
			126	"	"		"		"		B14	"	20 mA			"	0.8 V	"				"		"	29		"	
			127	"	"		"		"		B35	"	"	20 mA	0.8 V	"	"	"				"		"	30		"	
			128	"	"		"		"		B7	"	"	"	"	"	"	20 mA				"		"	41		"	
			129	"	"		"		"		"	"	"	"	"	"	"	"				"		"	42		"	
			130	"	"		"		"		"	"	"	"	"	"	"	"				"		"	43		"	
			131	"	"		"		"		"	"	"	"	"	"	"	"				"		"	44		"	
			132	"	"		"		"		"	"	"	"	"	"	"	"				"		"	45		"	
			133	"	"		"		"		"	"	"	"	"	"	"	"				"		"	46		"	
			134	"	"		"		"		"	"	"	"	"	"	"	"				"		"	47		"	
			135	"	"		"		"		"	"	"	"	"	"	"	"				"		"	48		"	
			136	"	"		"		"		"	"	"	"	"	"	"	"				"		"	49		"	
			137	"	"		"		"		"	"	"	"	"	"	"	"				"		"	50		"	
			138	"	"		"		"		"	"	"	"	"	"	"	"				"		"	51		"	
			139	"	"		"		"		"	"	"	"	"	"	"	"				"		"	52		"	
			140	"	"		"		"		"	"	"	"	"	"	"	"				"		"	53		"	
			141	"	"		"		"		"	"	"	"	"	"	"	"				"		"	54		"	
			142	"	"		"		"		"	"	"	"	"	"	"	"				"		"	55		"	
			143	"	"		"		"		"	"	"	"	"	"	"	"				"		"	56		"	
			144	"	"		"		"		"	"	"	"	"	"	"	"		20 mA					59		"	
			145	"	"		"		"		B35	"	"	"	"	"	"	"			20 mA				60		"	
			146	"	"		2.0 V		"		B14	"	"	"	"	2.0 V	"	B				2.0 V		2.0 V	61		"	
			147	"	"		2.0 V		"		B14	"	"	"	"	2.0 V	"	"				2.0 V		2.0 V	63		"	
	I_{IH}	3010	148	"	"		2.4 V																		4		300	µA
			149	"	"		"		2.4 V																6		300	µA
			150	"	"		"		"		2.4 V														8		1.0	mA
			151	"	"		"		"		"				2.4 V										31		300	µA
			152	"	"		"		"		"					2.4 V									32		"	"
			153	"	"		"		"		"					"	2.4 V								33		"	"
			154	"	"		"		"		"					"	"								34		"	"
			155	"	"		"		"		"					"	"								35		"	"
			156	"	"		"		"		"					"	"								36		"	"
			157	"	"		2.0 V		B		B35					2.0 V	B	2.4 V				2.0 V		2.0 V	41		"	"
			158	"	"		"		"		"					"	"	"				"		"	42		"	"
			159	"	"		"		"		"					"	"	"				"		"	43		"	"
			160	"	"		"		"		"					"	"	"				"		"	44		"	"
			161	"	"		"		"		"					"	"	"				"		"	45		"	"
			162	"	"		"		"		"					"	"	"				"		"	46		"	"
			163	"	"		"		"		"					"	"	"				"		"	47		"	"
			164	"	"		"		"		"					"	"	"				"		"	48		"	"
			165	"	"		"		"		"					"	"	"				"		"	49		"	"
			166	"	"		"		"		"					"	"	"				"		"	50		"	"
			167	"	"		"		"		"					"	"	"				"		"	51		"	"

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.
Terminals conditions (pins not designated are open) 1/

Subgroup	Symbol	MIL-STD-883 method	Test No.	1, 2, 27, 28	9, 26, 40, 57	3	4	5	6	7	8	10-24	29	30	31	32	33-36	41-56	59	60	61	62	63	64	Measured terminal	Test limits		Unit
				GND	INJ	WAIT	CTRD	HOLDA	RESET	IAQ	CLOCK	A14-A0	DBIN	CRUOUT	CRUIN	TRTREQ	IC3-IC0	DO-D15	CYCLE	CRUCLK	WE	READY	RCMEN	HOLD		Min	Max	
1 T_C = +25°C	I_{IH}	3010	168	GND	500 mA		2.0 V		B		335					2.0 V		B				2.0 V		2.0 V	52		300	µA
		"	169	"	"		"		"		"					"		"				"		"	53		"	"
		"	170	"	"		"		"		"					"		"				"		"	54		"	"
		"	171	"	"		"		"		"					"		"				"		"	55		"	"
		"	172	"	"		"		"		"					"		"				"		"	56		"	"
		"	173	"	"		"		"		"					"		"				2.4 V			62		"	"
		"	174	"	"		"		"		"					"		"				"		2.4 V	64		"	"
2	Same tests, terminal conditions, and limits as subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																											
3	Same tests, terminal conditions, and limits as subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																											
7, 8, 9, 10, and 11	FUNCTIONAL AND SWITCHING TESTS (not printed herein, see 4.5.2 and 6.4).																											

1/ "A" or "B" specifies the applicable waveform of figure 9; a numeral following the "A" or "B" indicates the number of clock pulses to be applied.

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3.5 Electrical test requirements. Electrical test requirements shall be as specified in table III for the applicable device type and device class. The subgroups of table III which constitute the minimum electrical test requirements for screening, and quality conformance by device class are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. The "JAN" or "J" certification mark shall not be used.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in a microcircuit group number covering I^2L LSI.

3.8 Manufacturer eligibility. To be eligible to supply microcircuits to this specification, a manufacturer shall have manufacturer certification in accordance with MIL-M-38510 for at least one line. Not necessarily the line producing the device type described herein.

3.9 Certification. Certification in accordance with MIL-M-38510 is not required for this device.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Qualification inspection. Qualification inspection is not required.

4.2.1 Combined LTPD's. During non-end point electrical parameter testing, LTPD's for subgroups shall be combined as follows:

Subgroups	Combined LTPD
1, 7, 9	5
2, 8, 10	7
3, 8, 11	7

4.3 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883 and as modified herein, and shall be conducted on all devices prior to quality conformance inspection.

a. The alternate screening option of 3.3 of MIL-STD-883, method 5004, is applicable.

b. Constant acceleration shall be 5,000 G maximum.

c. Burn-in test (method 1015 of MIL-STD-883).

1. Test condition D using the circuits shown on figures 6 and 7.

2. T_A = 125°C minimum.

d. Reverse bias burn-in and interim electrical test in accordance with 3.1.10 of method 5004 of MIL-STD-883 may be omitted.

e. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

f. The percent defective allowable (PDA) is specified as 10 percent for class B devices based on failures from group A, subgroups 1, 7, and 9 combined tests after cooldown as final electrical tests in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If

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interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroups 1, 7, and 9, after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.3). Generic test data (see 6.6) may be used to satisfy the requirements for groups C and D inspection. Quality conformance inspection shall be completed on the specific devices covered by this specification before they are shipped.

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

c. Variables data shall be taken for subgroups 9, 10, and 11 only after process or design changes.

d. LTPD's for subgroups shall be combined as follows:

Subgroups	Combined LTPD
1, 7, 9	7
2, 8, 10	10
3, 8, 11	10

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883.

4.4.3 Groups C and D inspections. Groups C and D inspections shall be in accordance with tables III and IV of method 5005 of MIL-STD-883 and as follows:

a. End point electrical parameters shall be as specified in table II herein.

b. Constant acceleration, subgroup 2 of group C and subgroup 4 of group D, shall be condition A (5,000 G).

c. Steady state life test (method 1005 of MIL-STD-883) conditions:

1. Test condition D, using the circuit shown on figure 6.

2. T_A = 125°C minimum.

3. Test duration - 1,000 hours, except as permitted in appendix B of MIL-M-38510.

4. Change LTPD to 10.

d. For moisture resistance and salt atmosphere of subgroups 3 and 5 of group D, omit initial conditioning.

4.5 Methods of examination and test. Methods of examination and test shall be as specified in the appropriate table and as follows:

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

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4.5.2 Functional and switching tests. Functional and switching tests (see 6.4) shall be performed using the pattern of table IV. The pattern definition and matrix key shall be as shown in figure 10. The timing for the pins under test (i.e., ABUS, DBUS, WE, etc.,) is given in figure 11. The following conditions shall apply:

		Functional test	AC test
Injector current		500 mA	500 mA
Inputs levels	V_{IH}	2.0 V	3.0 V
	V_{IL}	0.6 V	0.0 V
Output levels	V_{OH}	2.0 V	---
	V_{OL}	0.8 V	---
	Trip level	---	1.5 V
Frequency		1.0 MHz	3.0 MHz

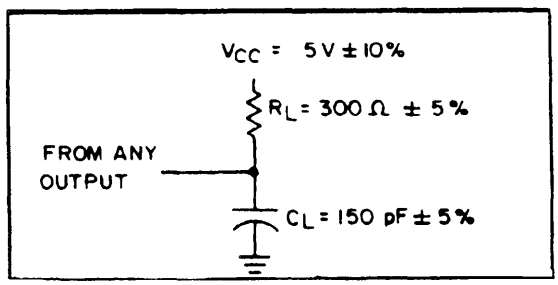


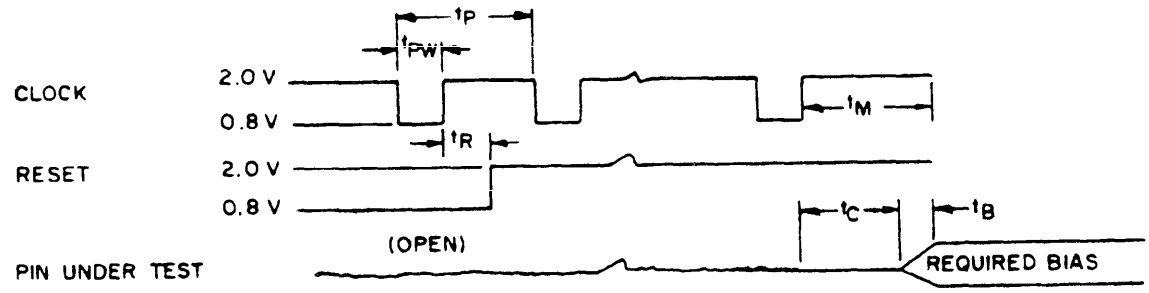
FIGURE 8. Switching times load circuits.

TABLE IV. Functional patterns for group A inspection (subgroups 7, 8, 9, 10, and 11).

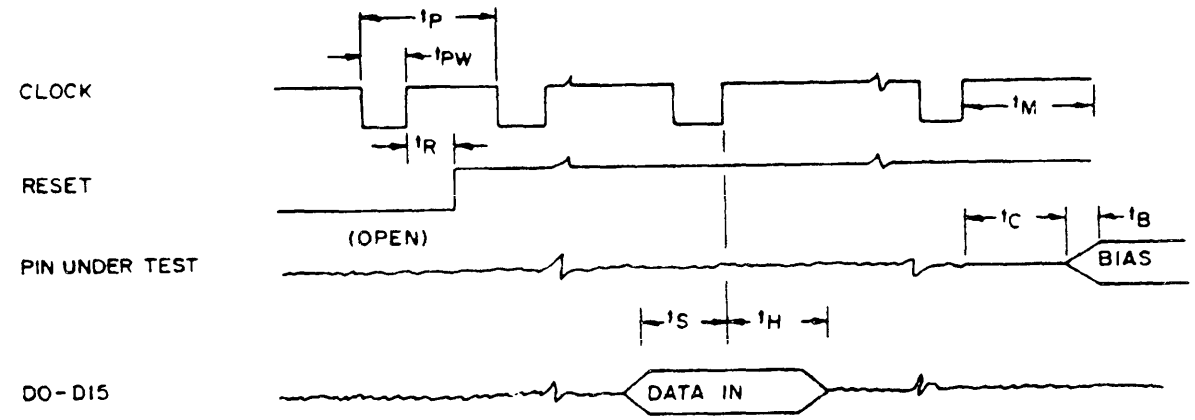
Pattern group	Pattern name	Number of tests
90107010	DC1N	291
90107100	INST1	1848
90107200	INST2	1922
90107300	INST3	1922
90107400	INST4	1864

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WAVEFORM A



WAVEFORM B



DO - D15 should have the following data in: V_{IH} = 2.0 V
V_{IL} = 0.8 V

Clock pulse	Data In (DO = MSD, D15 = LSD)
10,11	0000
26,27	19A0
30,31	1E7F
45,46	0000
61,62	19A0
65,66	1D7F

Definition of t terms

t_{PW} = Clock pulse width = 333 ns
t_P = Clock period = 1 µs
t_R = Reset hold time = 567 ns
t_S = Data setup time = 433 ns
t_H = Data hold time = 567 ns
t_B = Bias applied time = 500 µs
t_M = Total measurement time > 1.5 ms
t_C = Relay closure time = 1 ms

FIGURE 9. DC parametric waveforms.

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Functional patterns are defined in fields separated by commas. The fields are defined in hexadecimal with each bit corresponding to a device pin. The fields are filled from left to right with unused trailing bits set to zero. Field names are assigned as follows:

PTT	X'	B,	F,	0000,	FFFE,	0000,	FFFF,	10,	5,	0,	E,	0,	0,	C,	8;	0
PATTERN		CBUS	CBUSM	ABUS	ABUSM	DBUS	DBUSM	INT	ASYNI	ASYNO	ASYNOM	CRUI	CRUO	CRUOM	CLOCK	DRIVE

These fields correspond to the following pins:

CBUS	-	MEMEN, DBIN, WE, CYCEND
CBUSM	-	Mask for CBUS Pins
ABUS	-	A0 thru A14
ABUSM	-	Mask for ABUS Pins
DBUS	-	D0 thru D15
DBUSM	-	Mask for DBUS Pins
INT	-	IC0 thru IC3 and INTREQ
ASYNI	-	HOLD, READY, LOAD, RESET
ASYNO	-	HOLDA, WAIT, IAQ
ASYNOM	-	Mask for Asyno Pins
CRUI	-	CRUIN
CRUO	-	CRUCLK and CRUOUT
CRUOM	-	Mask for CRUO Pins
CLOCK	-	CLOCK Pin
DRIVE	-	Indicates direction for I/O Pins "1" indicates data being driven into part.

FIGURE 10. Functional patterns definition.

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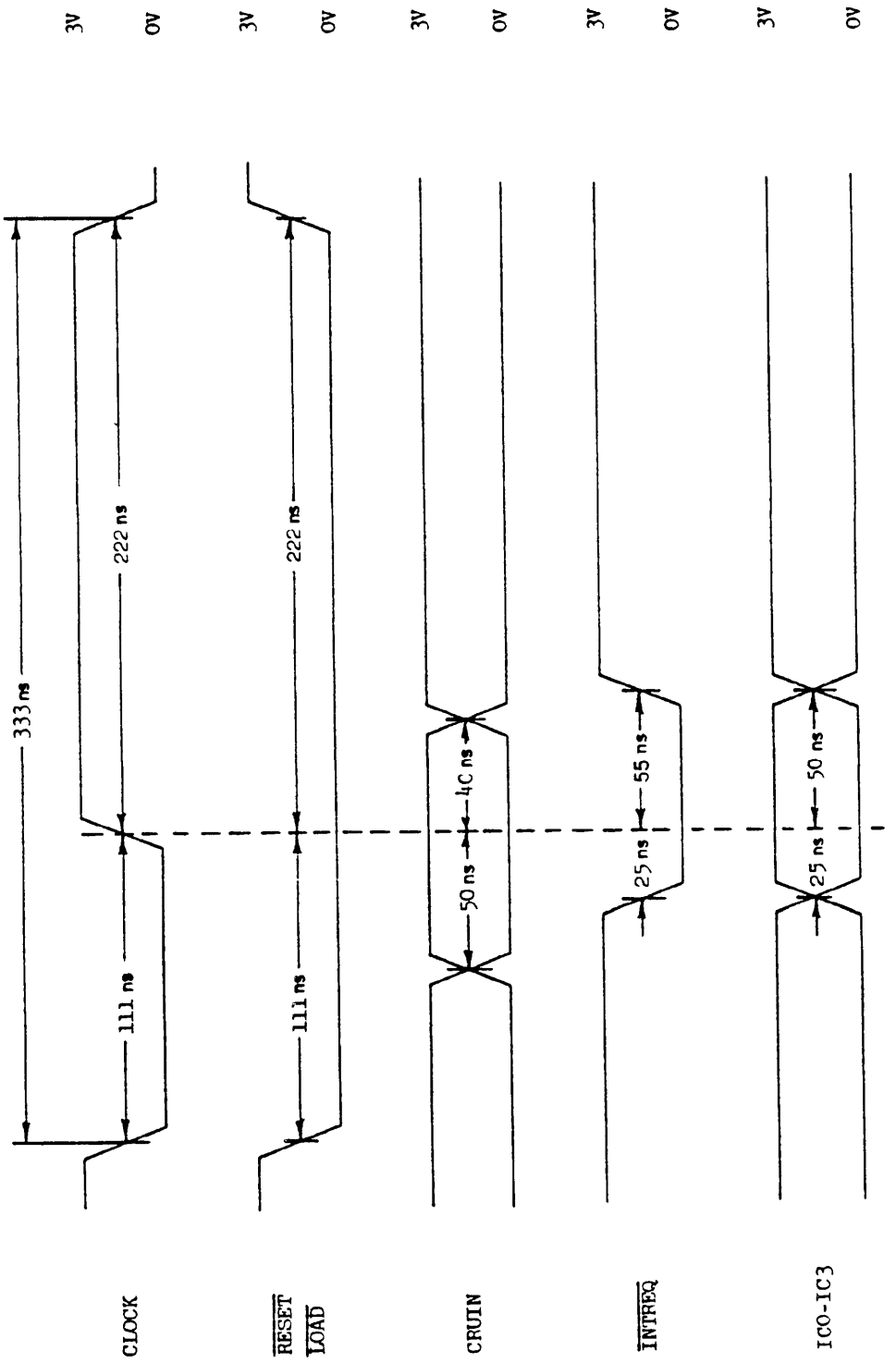
RPT 3X'	B, F,	000, FFFE,	0000, FFFF,	10,	5,	0, E,	0,	0, C,	8;	0
PATTERN (PTT)	EXPECTED	EXPECTED	EXPECTED	INPUT	INPUT	EXPECTED	INPUT	EXPECTED	C	D
OR	OUTPUT AND	OUTPUT AND	OUTPUT OR	DATA	DATA	OUTPUT AND	DATA	OUTPUT AND	L	R
REPEAT (RPT) AND COUNT	MASK FIELD FOR 'CBUS'	MASK FIELD FOR 'ABUS'	DRIVE DATA AND MASK FIELD FOR 'DBUS'	FOR 'INT'	FOR 'ASYNI'	MASK FIELD FOR 'ASYNO'	FOR 'CRUI'	MASK FIELD FOR 'CRUO'	O C K	I V E
										B I T

e.g. 'CBUS' = B, F = 1011, 1111 → MEMEN IS HIGH AND MASKED
DBIN IS LOW AND MASKED
WE IS HIGH AND MASKED
CYCEND IS HIGH AND MASKED

NOTES: 1. Drive bit indicates direction of DBUS.
'1' means data is input to 9900.
'0' means data is output from 9900.
2. 'ASYNI' input pattern data is inverted.

FIGURE 10. Functional patterns definition - Continued.

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(See figure 8 for load circuit)

FIGURE 11. AC parametric waveforms.

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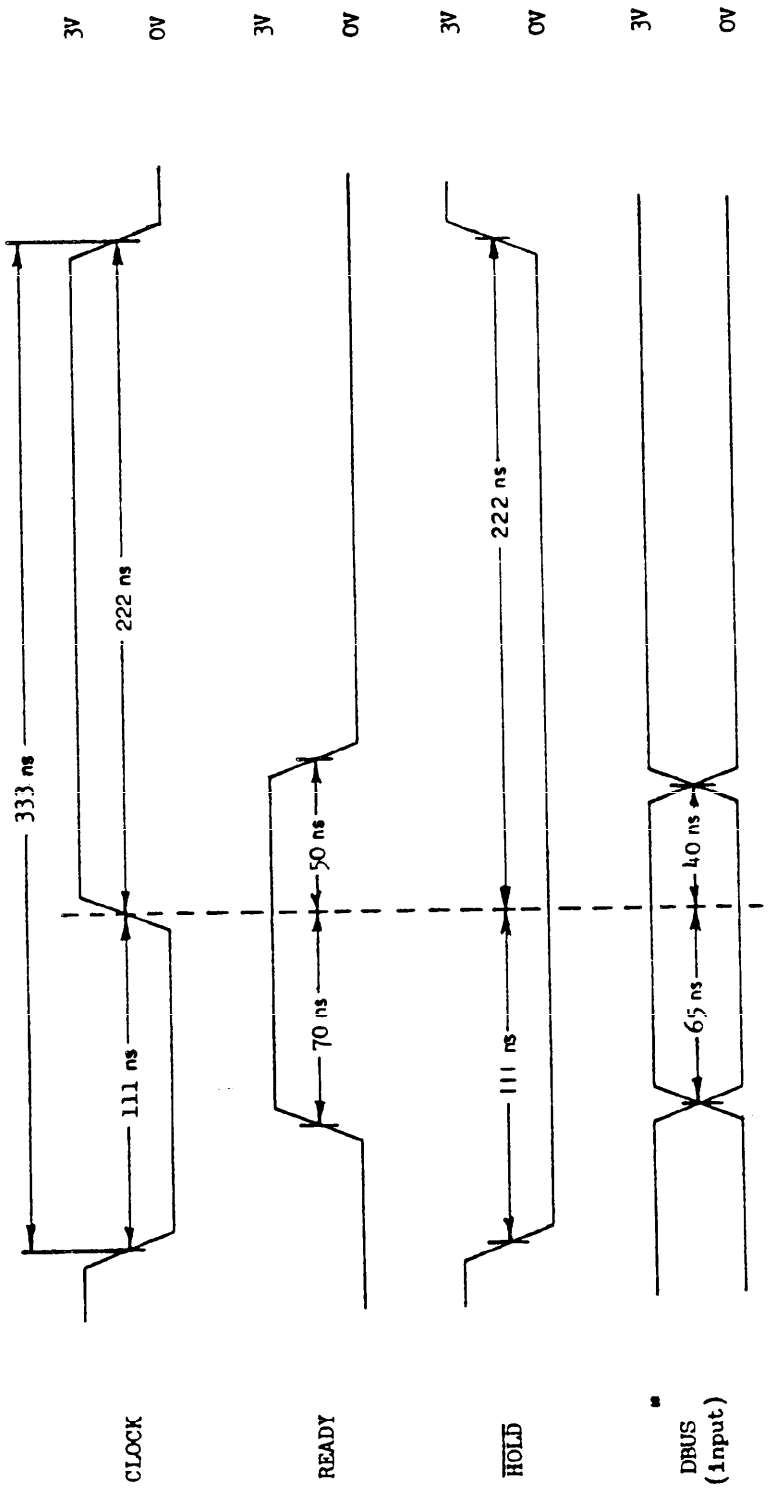


FIGURE 11. AC parametric waveforms - Continued.

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identified by a single letter in accordance with table I.

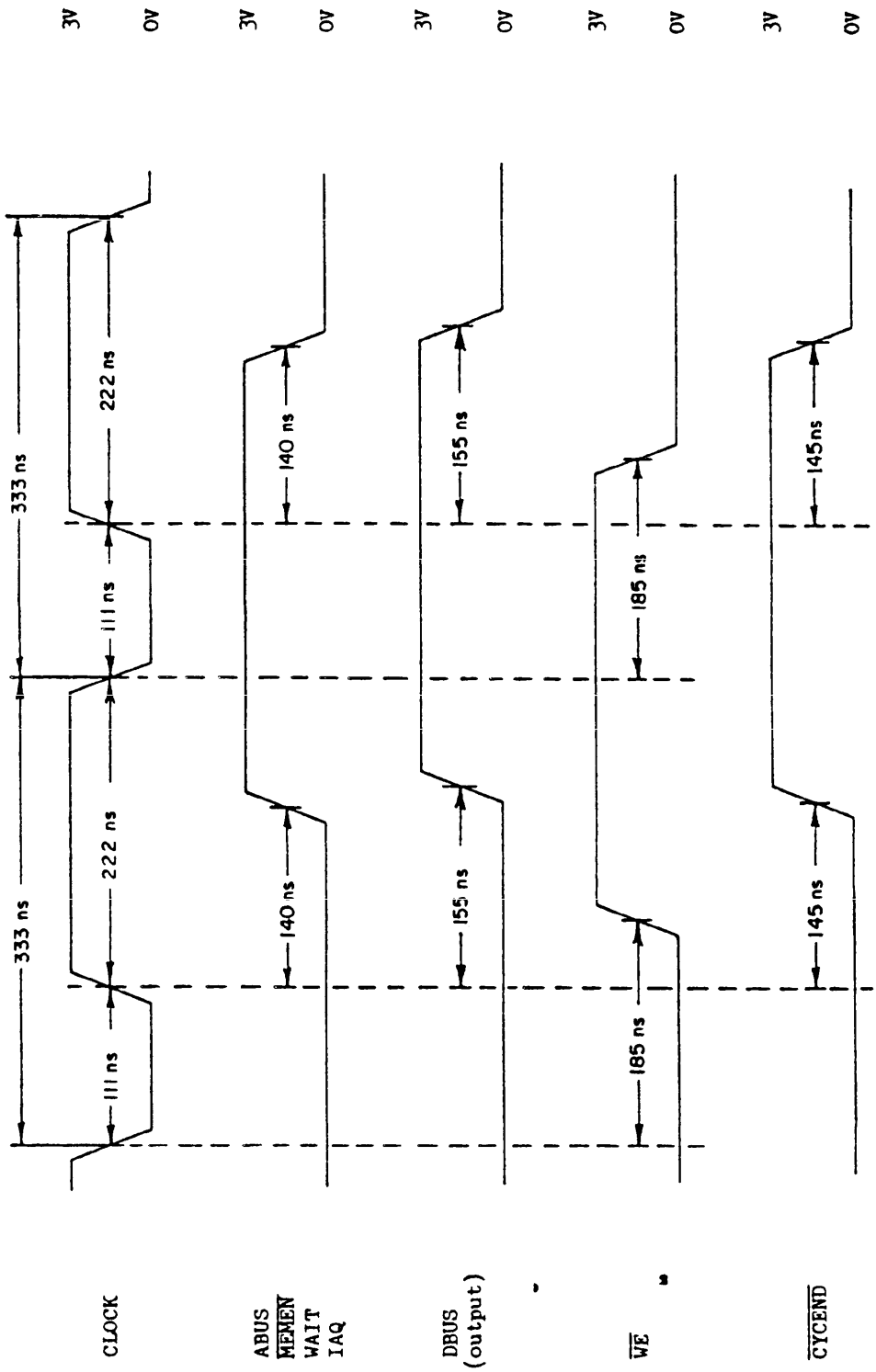


FIGURE 11. AC parametric waveforms - Continued

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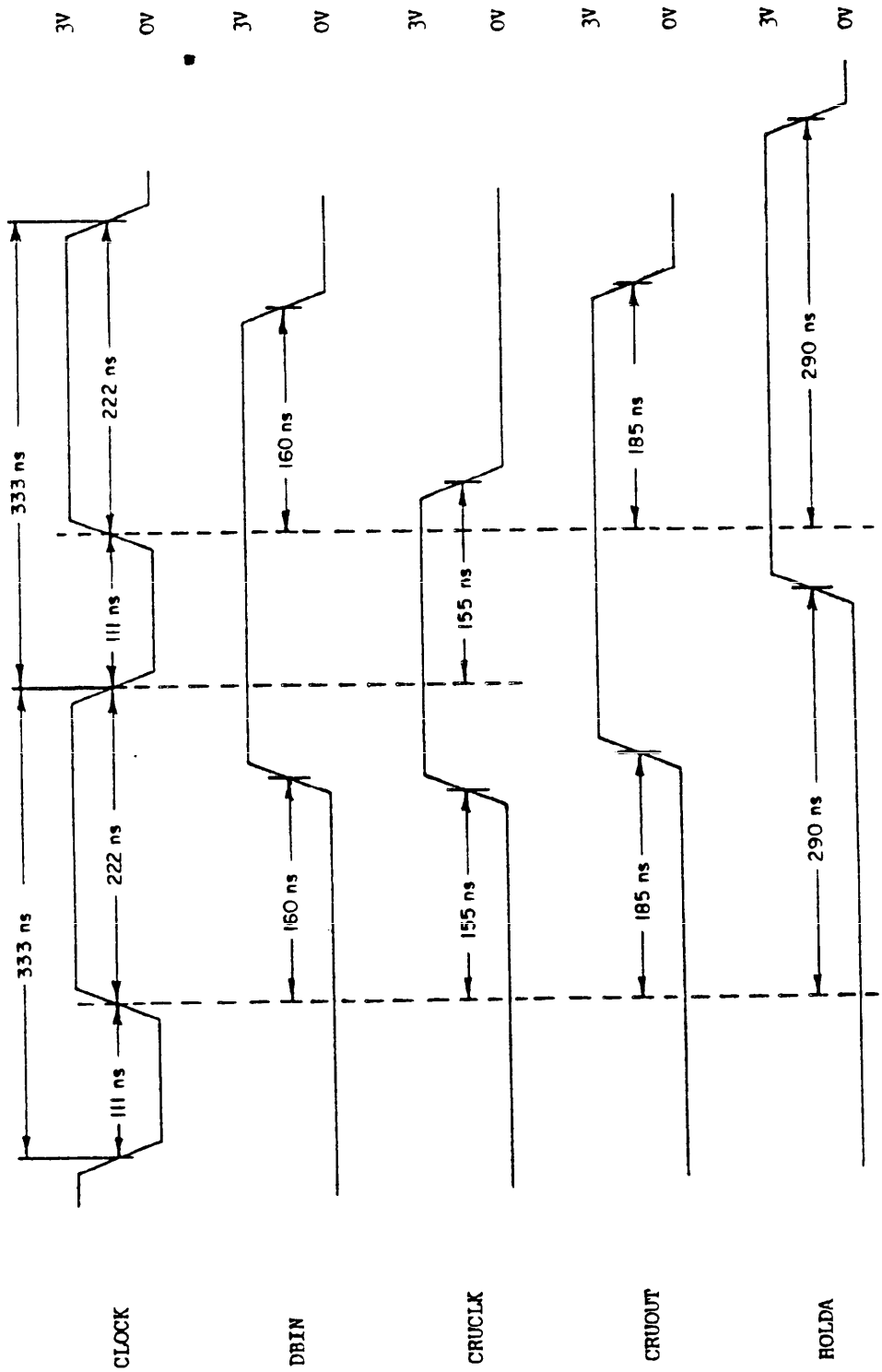


FIGURE 11. AC parametric waveforms - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirement for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for special carriers, lead lengths or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331, and as follows:

GND	Electrical ground (common terminal)
V_{IN}	Voltage level at an input terminal
V_{IC}	Input clamp voltage
I_{IN}	Current flowing into an input terminal

6.3.1 Pin definitions. The following describes the function of each pin (see figure 2 for their assigned location):

Signature	Pin No(s)	I/O	Description
			ADDRESS BUS
A0(MSB) - A14(LSB)	24 - 10	OUT	A0(MSB) through A14 (LSB) comprise the address bus. This open-collector bus provides the memory-address vector to the external-memory system when $\overline{\text{MEMEN}}$ is active, and I/O-bit addresses to the I/O system when $\overline{\text{MEMEN}}$ is inactive. When HOLDA is active, the address bus is pulled to the logic level HIGH state by the individual pull-up resistors tied to each respective open-collector output.

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Signature	Pin No(s)	I/O	Description
			DATA BUS
D0(MSB) - D15(LSB)	41 - 56	I/O	D0(MSB) through D15(LSB) comprise the bid-directional open-collector data bus. This bus transfers memory data to (when writing) and from (when reading) the external memory system when $\overline{\text{MEMEN}}$ is active. When HOLDA is active, the data bus is pulled to the logic level HIGH state by the individual pull-up resistors tied to each respective open-collector output.
			POWER SUPPLY
INJ	9,26,40,57	---	Injector-supply-current
GND	1,2,27,28	---	Ground reference
			CLOCK
CLOCK	8	IN	CLOCK
			BUS CONTROL
DBIN	29	OUT	DATA BUS IN. When active (pulled to logic level HIGH), DBIN indicates that the device has disabled its output buffers to allow the memory to place memory-read data on the data bus during $\overline{\text{MEMEN}}$. DBIN remains at logic level LOW in all other cases except when HOLDA is active (pulled to logic level HIGH).
$\overline{\text{MEMEN}}$	63	OUT	MEMORY ENABLE. When active (logic level LOW, $\overline{\text{MEMEN}}$ indicates that the address bus contains a memory address.
$\overline{\text{WE}}$	61	OUT	WRITE ENABLE. When active (logic level LOW, $\overline{\text{WE}}$ indicates that the data bus is outputting data to be written into memory.
CRUCLK	60	OUT	COMMUNICATIONS-REGISTER-UNIT (CRU) CLOCK. When active (pulled to logic level HIGH), CRUCLK indicates to the external interface logic the presence of output data on CRUOUT, or the presence of an encoded external instruction on A0 through A2.
CRUIN	31	IN	CRU DATA IN. CRUIN, normally driven by 3-state or open-collector devices, receives input data from the external interface logic. When the device executes a STCR or TB instruction, it samples CRUIN for the level of the CRU input bit specified by the address bus (A3 through A14).

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t allowed) (Note: Solder dip of gold plated leads is no

Signature	Pin No(s)	I/O	Description
CRUOUT	30	OUT	CRU DATA OUT. CRUOUT outputs single-bit data when the device executes a LDCR, SBZ, SBO instruction. The data on CRUOUT should be sampled by the external interface logic when CRUCLK goes active (pulled to logic level HIGH).
			INTERRUPT CONTROL
$\overline{\text{INTREQ}}$	32	IN	INTERRUPT REQUEST. When active (logic level LOW), INTREQ indicates that an external interrupt is requesting service. If $\overline{\text{INTREQ}}$ is active, the device loads the data on the interrupt-code input lines IC0 through IC3 into the internal interrupt-code storage register. The code is then compared to the interrupt mask bits of the status register. If equal or higher priority then the enabled interrupt level (interrupt code equal or less than status register bits 12 through 15), the device initiates the interrupt sequence. If the comparison fails, the device ignores the interrupt request. In that case, $\overline{\text{INTREQ}}$ should be held active. The device will continue to sample IC0 through IC3 until the program enables a sufficiently low interrupt-level to accept the requesting interrupt.
IC0 (MSB) - IC3 (LSB)	36 - 33	IN	INTERRUPT CODES. IC0 (MSB) through IC3 (LSB), receiving an interrupt identify code, are sampled by the device when $\overline{\text{INTREQ}}$ is active (logic level LOW). When IC0 through IC3 are LLLH, the highest priority external interrupt is requesting service; when HHHH, the lowest priority external interrupt is requesting service.
			MEMORY CONTROL
$\overline{\text{HOLD}}$	64	IN	When active (logic level LOW), $\overline{\text{HOLD}}$ indicates to the device that an external controller (e.g., DMA device) desires to use both the address bus and data bus to transfer data to or from memory. In response, the device enters the hold state after completion of its present memory cycle. The device then allows its address bus, data bus, $\overline{\text{WE}}$, $\overline{\text{MEMEN}}$, DBIN, and HOLDA facilities to be pulled to the logic level HIGH state. When $\overline{\text{HOLD}}$ is deactivated, the device returns to normal operation from the point at which it was stopped. $\overline{\text{HOLD}}$ must be applied and released synchronously with the falling edge of CLOCK.

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Signature	Pin No(s)	I/O	Description
HOLDA	5	OUT	HOLD ACKNOWLEDGE. When active (pulled to logic level HIGH), HOLDA indicates that the device is in the hold state and that its address bus, data bus, $\overline{WE}$, $\overline{MEMEN}$, and DBIN facilities are pulled to the logic level HIGH state.
READY	62	IN	When active (logic level HIGH), READY indicates that the memory will be ready to read or write during the next clock cycle. When not-ready is indicated during a memory operation, the device enters a wait state and suspends internal operation until the memory systems activate READY.
WAIT	3	OUT	When active (pulled to logic level HIGH), WAIT indicates that the device has entered a wait state in response to a not-ready condition from memory.
			TIMING AND CONTROL
IAQ	7	IN	INSTRUCTION ACQUISITION. IAQ is active (pulled to logic level HIGH) during any device initiated instruction acquisition memory cycle. Consequently, IAQ may be used to facilitate detection of illegal op codes.
$\overline{CYCEND}$	59	OUT	CYCLE END. When active (logic level LOW), $\overline{CYCEND}$ indicates that the device will initiate a new microinstruction cycle on the low-to-high transition of the next CLOCK.
$\overline{LOAD}$	4	IN	When active (logic level LOW), $\overline{LOAD}$ causes the device to execute a nonmaskable interrupt with memory addresses $FFFC_{16}$ and $FFFE_{16}$ containing the associated trap vectors (WP and PC). The load sequence is initiated after the instruction being executed is completed. $\overline{LOAD}$ will also terminate an idle state. If $\overline{LOAD}$ is active during the time $\overline{RESET}$ is active, the $\overline{LOAD}$ trap will occur after the $\overline{RESET}$ function is completed. $\overline{LOAD}$ should remain active for one instruction execution period (IAQ may be used to monitor instruction boundaries).
$\overline{RESET}$	6	IN	When active (logic level LOW), $\overline{RESET}$ causes the device to reset itself and inhibit $\overline{WE}$ and CRUCLK. When $\overline{RESET}$ is released, the device initiates a level-zero interrupt sequence acquiring the WP and PC trap vectors from memory locations 0000_{16} and 0002_{16}, sets all status register bits to logic level LOW, and then fetches the first instruction of the reset program environment. RESET must be held active for a minimum of three CLOCK cycles.

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6.4 Functional pattern sets. The table for the functional and switching tests is not printed herein because of its extreme length and pattern complexity. The test program output should be obtained in a form compatible with test system architecture (magnetic tape format usually preferred). The preparing activity (or DESC, as its agent) may be consulted for information on obtaining such output.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Generic test data. Generic test data may be used to satisfy the requirements of 4.4.3. Group C generic test data shall be on date codes no more than one year old and on a die in the same microcircuit group (see appendix E of MIL-M-38510) with the same material, design and process and from the same plant as the die represented. Group D (see 4.4.3) generic data shall be on date codes no more than one year old and on the same package type (see terms, definitions, and symbols of MIL-M-38510) and from the same plant as the package represented. The vendor is required to retain the generic data for a period of not less than 36 months from the date of shipment.

6.7 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliabilty factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	9900A

6.8 Ordering guidance. Since the qualification requirements have been removed from the specification, orders may be placed immediately.

6.9 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 17

Review activities:
Air Force - 11, 19, 85, 99
DLA - ES

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-F667-1)

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APPENDIX

GENERAL FEATURES, FUNCTIONAL DESCRIPTION, AND INSTRUCTION SET FOR 16-BIT MICROPROCESSOR

10. SCOPE

10.1 This appendix covers the general features, functional description, and instruction set for the parallel 16-bit microprocessor covered by this specification.

20. GENERAL FEATURES

20.1 Interfacing. The input/output (I/O) accommodations have been designed for TTL compatibility. Direct interfacing, supportable by the entire families of support devices, is shown in figure 12.

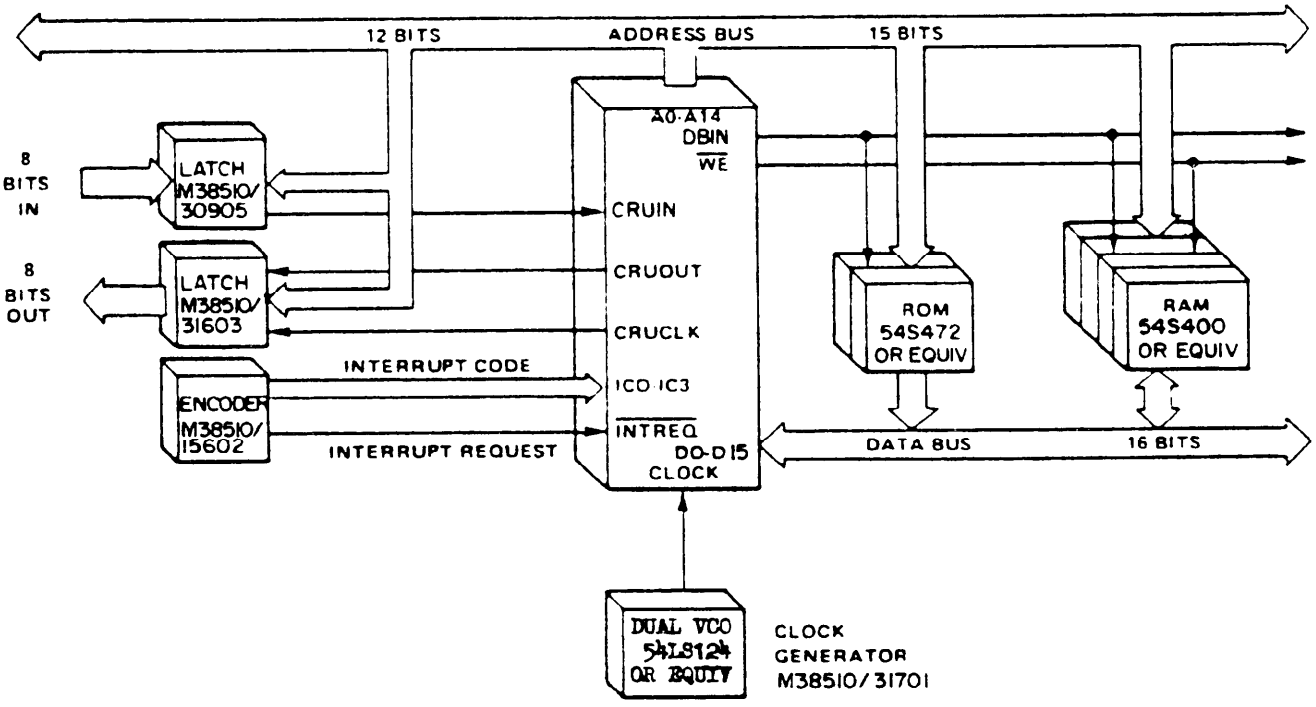


FIGURE 12. Microprocessor and support devices.

20.1.1 Input circuit. The input circuit used is basically an RTL configuration which has been modified for TTL compatibility as shown in figure 13A. An input-clamping diode is incorporated to limit negative excursions (ringing) when the device is on the receiving end of a transmission line; an input switching threshold of nominally +1.5 volts has been specified for improved noise immunity. This threshold is achieved via two resistors which function as a voltage divider to increase the one V_{BE} threshold of the I^2L input transistor to +1.5 volts. Since this input circuit is independent of injector current, input threshold compatibility is maintained over the entire speed/power performance range.

The input circuit characteristics for input current versus input voltage are shown in figure 14. The 10K and 20K ohm load lines and threshold knee at +1.5 volts provide a high-impedance characteristic to reduce input loading and improve the low-logic level input noise immunity over some standard TTL inputs. Full compatibility is maintained with virtually all 5 volt logic families even when the device is powered down (injector current reduced).

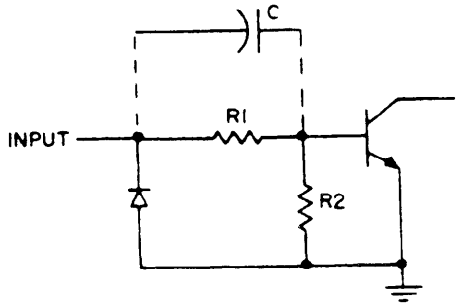
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20.1.1.1 Sourcing inputs. The inputs may be sourced directly by most 5 volt logic families. Five volt functions which feature internal pull-up resistors at their outputs require no external interface components; 5 volt functions which feature open-collector outputs generally require external pull-up resistors.

20.1.1.2 Terminating unused inputs. Inputs which are selected to be hardwired to a logic-level low may be connected directly to ground. Inputs which are selected to be hardwired to a logic-level high may be connected directly to +5 volts.

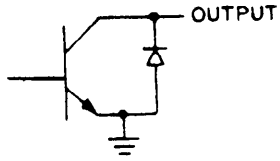
20.1.2 Output circuit. The output circuit selected is an injected open-collector transistor shown in figure 13B. Since this transistor is injected, output sourcing capability is directly related to injector current. In other words, the number of loads which may be sourced by an output is directly reduced as injector current is reduced.

13A. Equivalent of each input.

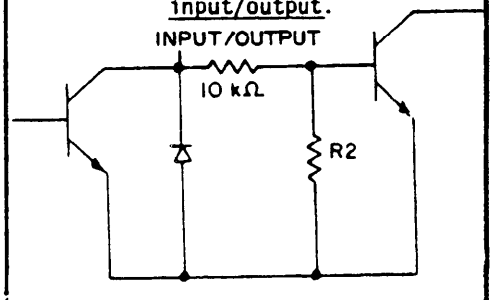


CLOCK: R1 = 3 kΩ, R2 = 10 kΩ
RESET: R1 = 10 kΩ, R2 = 10 kΩ
CRUIN, READY: R1 = 10 kΩ, R2 = 13 kΩ
All others: R1 = 10 kΩ, R2 = 11 kΩ
*Cap across 3 kΩ res, C = 5pF

13B. Typical of all outputs.



13C. Equivalent of each input/output.



D7 - D11: R2 = 13 kΩ
All others: R2 = 10 kΩ

FIGURE 13. Schematics of equivalent inputs, outputs, inputs/ outputs.

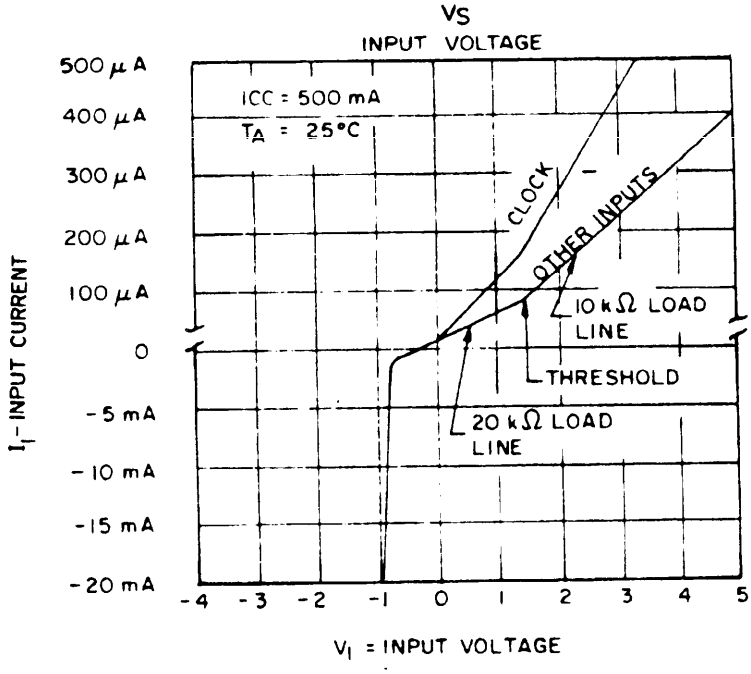


FIGURE 14. Typical input characteristics.

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20.2 Architecture. The architecture (block diagram) and microprocessor flow chart are as specified in 3.2.2. The memory word length is 16 bits long. Each word is also defined as 2 bytes of 8 bits. The instruction set allows both word and byte operands. Thus, all memory locations are on even address boundaries and byte instructions can address either the even or odd byte. The memory space is 65,536 bytes or 32,768 words. The word and byte formats are shown on figure 15.

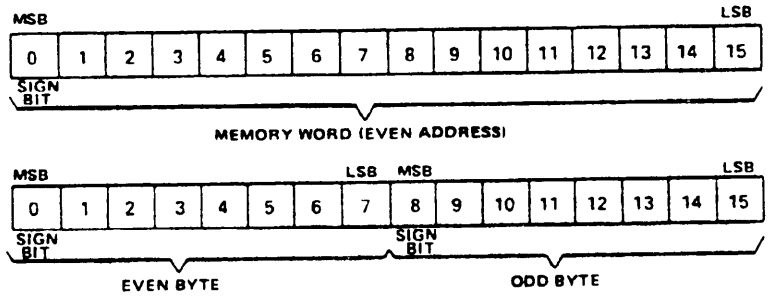


FIGURE 15. Word and byte formats.

30. FUNCTIONAL DESCRIPTION

30.1 Registers amd memory. The device employs an advanced memory-to-memory architecture. Blocks of memory designated as workspace replace internal hardware registers with program-data registers. The device memory map is shown on figure 5. The first 32 words are used for interrupt trap vectors. The next contiguous block of 32 memory words is used by the extended operation (XOP) instruction for trap vectors. The last two memory words, $FFFC_{16}$ and $FFFE_{16}$, are used for the trap vector of the $\overline{LOAD}$ signal. The remaining memory is then available for programs, data, and workspace registers. If desired, any of the special areas may also be used as general memory.

Three internal registers are accessible to the user. The program counter (PC) contains the address of the instruction following the current instruction being executed. This address is referenced by the processor to fetch the next instruction from memory and is then automatically incremented. The status register (ST) contains the present state of the processor and will be further defined in 60.4. The workspace pointer (WP) contains the address of the first word in the currently active set of workspace registers.

A workspace-register file occupies 16 contiguous memory words in the general memory area (see figure 5). Each workspace register may hold data or addresses and function as operand registers, accumulators, address registers, or index registers. During instruction execution, the processor addresses any register in the workspace by adding the register number to the contents of the workspace pointer and initiating a memory request for the word. The relationship between the workspace pointer and its corresponding workspace is shown in figure 16.

The workspace concept is particularly valuable during operations that require a context switch which is a change from one program environment to another (as in the case of an interrupt) or to a subroutine. Such an operation, using a conventional multi-register arrangement, requires that at least part of the contents of the register file be stored and reloaded. A memory cycle is required to store or fetch each word. By exchanging the program counter, status register, and workspace pointer in the device concept accomplishes a complete context switch with only three store cycles and two fetch cycles. See figure 16. After the switch, the workspace pointer contains the starting address of a new 16-word workspace in memory for use in the new routine. A corresponding time-saving occurs when the original context is restored. Instructions in the device that result in a context switch include:

a. Branch and load workspace pointer (BLWP).
b. Return from subroutine (RTWP).
c. Extended operation (XOP).

Device interrupts, $\overline{RESET}$, and $\overline{LOAD}$ also cause a context switch by forcing the processor to trap to a service subroutine.

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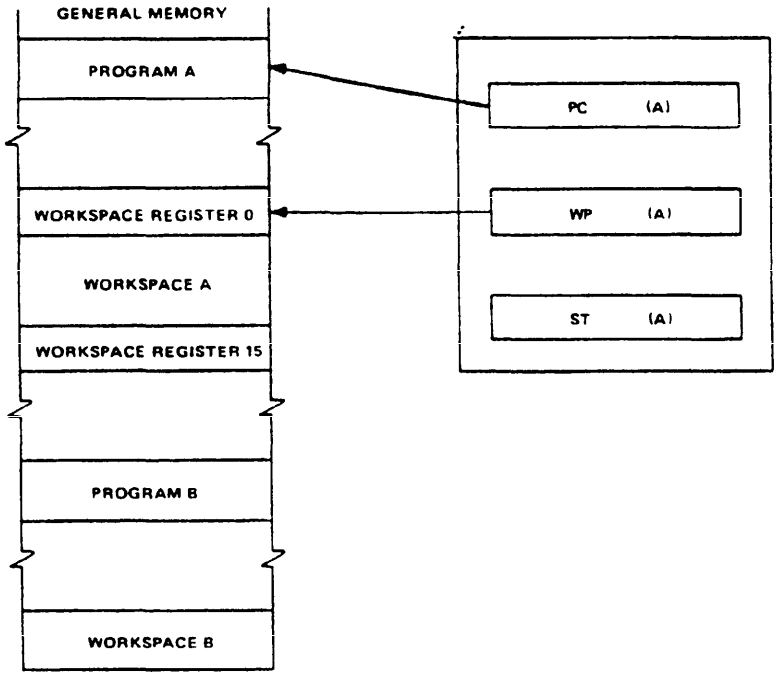


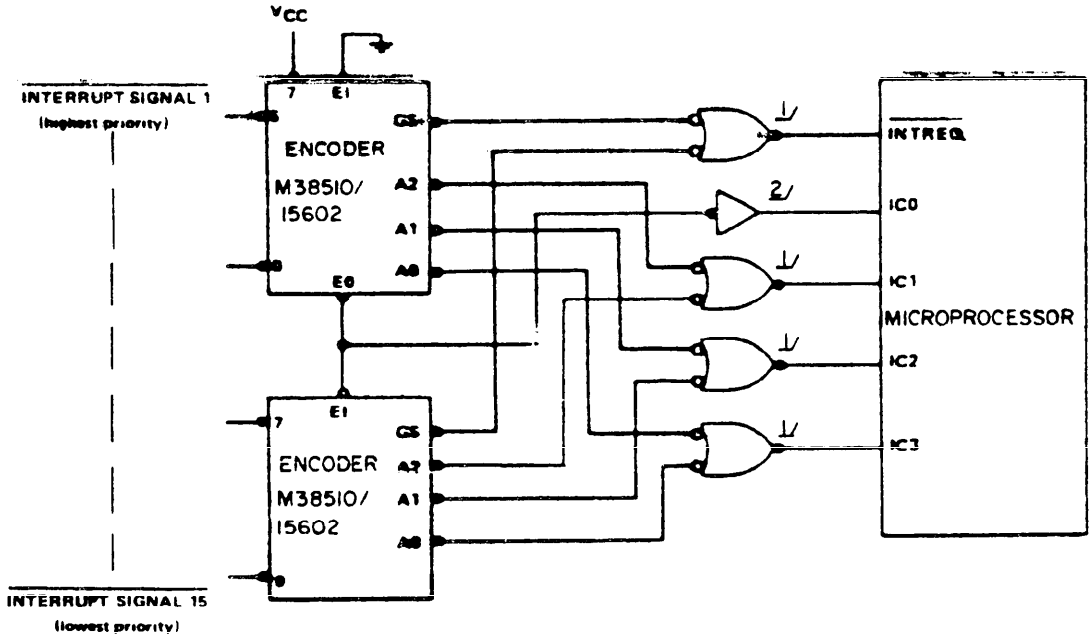
FIGURE 16. Memory-to-memory workspace concept.

30.2 Interrupts. The device employs 16 interrupt levels with the highest priority level 0 and lowest level 15. Level 0 is reserved for the RESET function and all other levels may be used for external devices. The external levels may be also shared by several device interrupts, depending upon system requirements. The device continuously compares the interrupt code (IC0 through IC3) with the interrupt mask contained in status-register bits 12 through 15. When the level of the pending interrupt is less than or equal to the enabling mask level (higher or equal priority interrupt), the processor recognizes the interrupt and initiates a context switch following completion of the currently executing instruction. The processor fetches the new context WP and PC from the interrupt vector locations. Then, the previous context WP, PC, and ST are stored in workspace registers 13, 14, and 15, respectively, of the new workspace. The device then forces the interrupt mask to a value that is one less than the level of the interrupt being serviced, except for level-zero interrupt, which loads zero into the mask. This allows only interrupts of higher priority to interrupt a service routine. The processor also inhibits interrupts until the first instruction of the service routine has been executed. All interrupt requests should remain active until recognized by the processor in the device-service routine. The individual service routines must reset the interrupt requests before the routine is complete.

If a higher priority interrupt occurs, a second context switch occurs to service the higher priority interrupt. When that routine is complete, a return instruction (RTWP) restores the first service routine parameters to the processor to complete processing of the lowest-priority interrupt. All interrupt subroutines should terminate with the return instruction to restore original program parameters. The interrupt-vector locations, device assignment, enabling-mask value, and the interrupt code are shown in table V.

The interrupt interface utilizes standard TTL components as shown in figure 17. Note that for eight or less external interrupts a single encoder is required and for one external interrupt INTREQ is used as the interrupt signal with a hardwired code IC0 through IC3.

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1/ M38510/01601
2/ M38510/00105

FIGURE 17. Interrupt interface.

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TABLE V. Interrupt level data.

Interrupt level		Vector location (memory address in Hex)	Device assignment	Interrupt mask values to enable respective interrupts (ST12 thru ST15)	Interrupt codes IC0 thru IC3
(Highest priority)	0	00	Reset	0 through F 1/	0000
	1	04	External device	1 through F	0001
	2	08	↓	2 through F	0010
	3	0C		3 through F	0011
	4	10		4 through F	0100
	5	14		5 through F	0101
	6	18		6 through F	0110
	7	1C		7 through F	0111
	8	20		8 through F	1000
	9	24		9 through F	1001
	10	28		A through F	1010
	11	2C		B through F	1011
	12	30		C through F	1100
	13	34		D through F	1101
	14	38		E and F	1110
(Lowest priority)	15	3C	External device	F only	1111

1/ Level 0 cannot be disabled.

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30.3 I/O interface communications-register-unit (CRU). The communications register-unit (CRU) is a versatile, direct command-driven, single-bit I/O interface. The CRU may directly address up to 4096 peripheral input bits and up to 4096 peripheral output bits. It executes three single-bit and two multiple-bit CRU instructions. The single-bit instructions are TEST BIT (TB), SET BIT TO ONE (SBO), and SET BIT TO ZERO (SBZ). The multiple-bit instructions are LOAD CRU (LDCR) and STORE CRU (STCR).

The device employs three dedicated I/O signals CRUIN, CRUOUT, CRUCLK, and the least significant twelve bits of the address bus to support the CRU interface. CRU interface timing is described in 40.3.

30.4 Single-bit CRU operations. The device performs three single-bit CRU functions: test bit (TB), set bit to one (SBO), and set bit to zero (SBZ). To identify the bit to be operated upon, the device develops a CRU-bit address and places it on the address bus, A3 to A14.

For the two output operations (SBO and SBZ), the processor also generates a CRUCLK pulse, indicating an output operation to the CRU device, and places bit 7 of the instruction word on the CRUOUT line to accomplish the specified operation (bit 7 is a one for SBO and a zero for SBZ). A test-bit instruction transfers the addressed CRU bit from the CRUIN input line to bit 2 of the status register (EQUAL).

The device develops a CRU-bit address for the single-bit operations from the CRU-base address contained in workspace register 12 and the signed displacement count contained in bits 8 through 15 of the instruction. The displacement allows two's complement addressing from base minus 128 bits through base plus 127 bits. The base address from W12 is added to the signed displacement specified in the instruction and the result is loaded onto the address bus. Figure 18 illustrates the development of a single-bit CRU address.

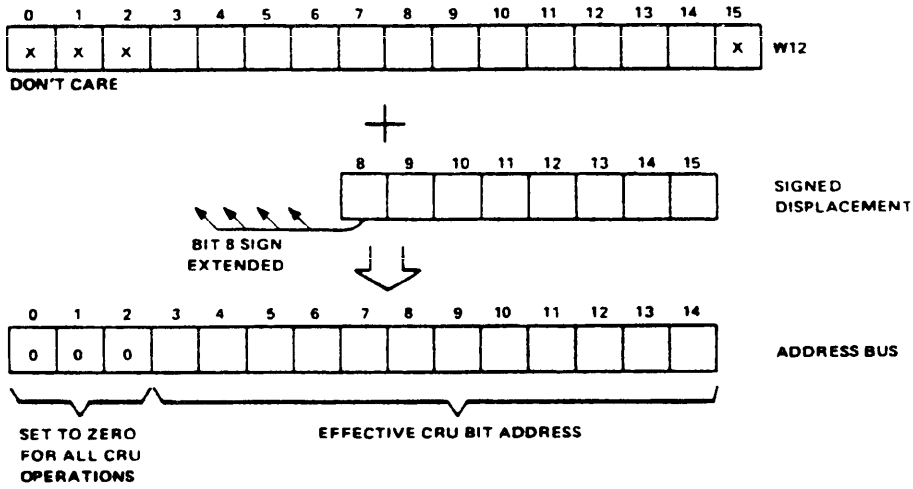


FIGURE 18. Single-bit CRU address development.

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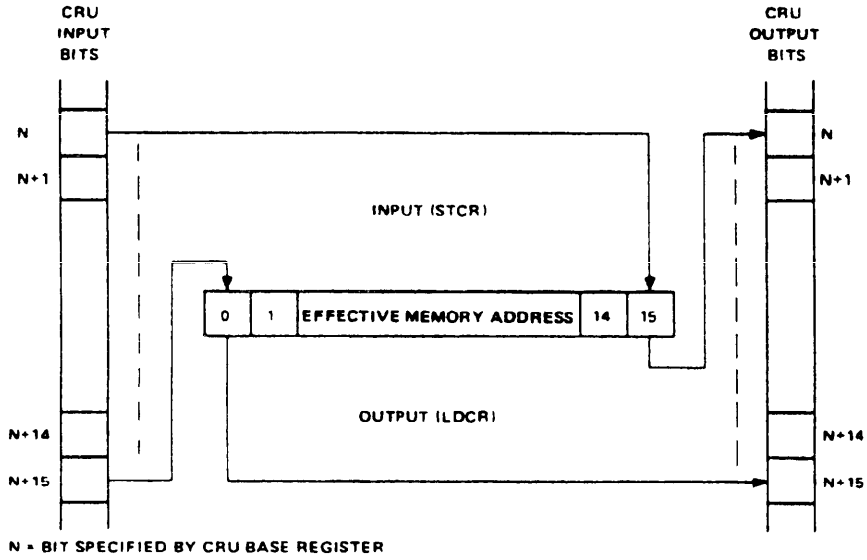


FIGURE 19. LDCR/STCR data transfers.

30.5 Multiple-bit CRU operations. The device performs two multiple-bit CRU operations: store communications register (STCR) and load communications register (LDCR). Both operations perform a data transfer from the CRU-to-memory or from memory-to-CRU as illustrated in figure 19. Although the figure illustrates a full 16-bit transfer operation, any number of bits from 1 through 16 may be involved. The LDCR instruction fetches a word from memory and right-shifts it to serially transfer it to CRU output bits. If the LDCR involves eight or fewer bits, those bits come from the right-justified field within the addressed byte of the memory word. If the LDCR involves nine or more bits, those bits come from the right-justified field within the whole memory word. When transferred to the CRU interface, each successive bit receives an address that is sequentially greater than the address for the previous bit. This addressing mechanism results in an order reversal of the bits; that is, bit 15 of the memory word (or bit 7) becomes the lowest addressed bit in the CRU and bit 0 becomes the highest addressed bit in the CRU field.

An STCR instruction transfers data from the CRU to memory. If the operation involves a byte or less transfer, the transferred data will be stored right-justified in the memory byte with leading bits set to zero. If the operation involves from 9 to 16 bits, the transferred data is stored right-justified in the memory word with leading bits set to zero. When the input from the CRU device is complete, the first bit from the CRU is the least-significant-bit position in the memory word or byte.

Figure 20 illustrates how to implement a 16-bit input and a 16-bit output register using the CRU interface. CRU addresses are decoded as needed to implement up to 256 such 16-bit interface registers. In system application, however, only the exact number of interface bits needed to interface specific peripheral devices are implemented. It is not necessary to have a 16-bit interface register to interface an 8-bit device.

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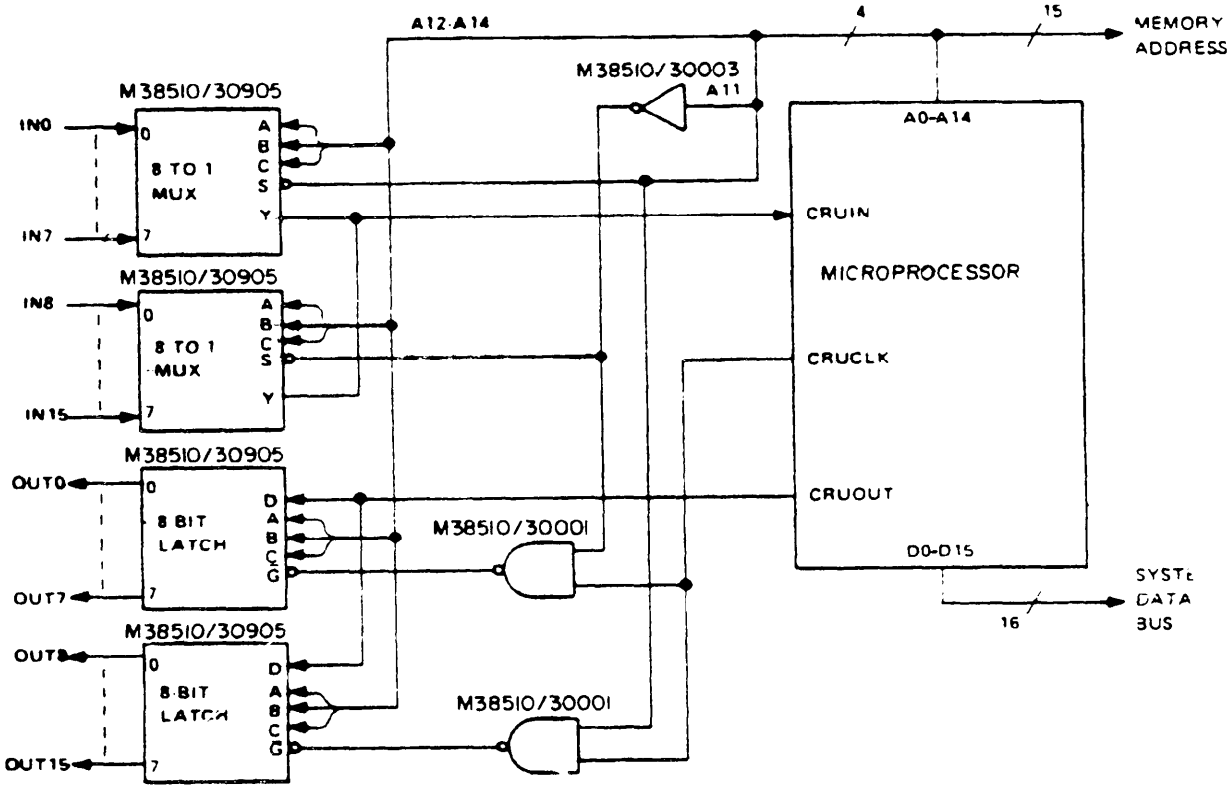


FIGURE 20. 16-BIT input/output interface.

30.6 External instructions. The device has five external instructions that allow user-defined external functions to be initiated under program control. These instructions are CKON, CKOF, RSET, IDLE, and LREX. IDLE also causes the device to enter the idle state and remain until an interrupt, RESET, or LOAD occurs. When any of these five instructions are executed by the device, a unique 3-bit code appears on the most significant 3 bits of the address bus (A0 through A2) along with a CRUCLK pulse. When the device is in an idle state, the 3-bit code and CRUCLK pulses occur repeatedly until the idle state is terminated. The codes are shown in table VI. Figure 21 illustrates typical external decode logic to implement these instructions. Note that a signal is generated to inhibit CRU decodes during external instruction.

TABLE VI. External instructions.

External instruction	A0	A1	A2
LREX	H	H	H
CKOF	H	H	L
CKON	H	L	H
RSET	L	H	H
IDLE	L	H	L

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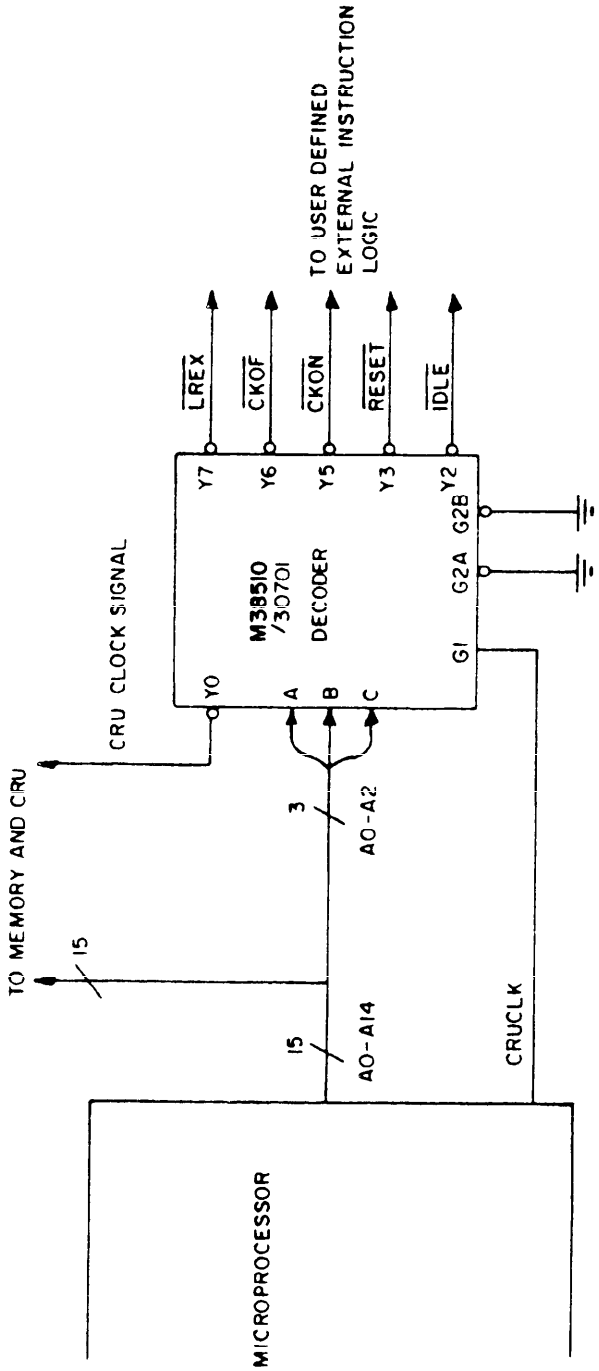


FIGURE 21. External instruction DECODE logic.

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30.7 Load function. The $\overline{\text{LOAD}}$ signal allows volatile memory to be used for the $\overline{\text{RESET}}$ trap vectors. When active, $\overline{\text{LOAD}}$ causes the device to initiate an interrupt sequence immediately following the instruction being executed. Memory locations FFFC and FFFE are used to obtain the trap vectors (WP and PC). The old PC, WP, and ST are loaded into the new workspace and the interrupt mask is set to 0000.

40. TIMING

40.1 Memory. The basic memory timing for a memory-read cycle with no wait states, and a memory-write cycle with one wait state, is shown in figure 22. During each memory-read or memory-write cycle, $\overline{\text{MEMEN}}$ becomes active (logic level LOW) along with valid memory-address data appearing on the address bus (A0 through A14).

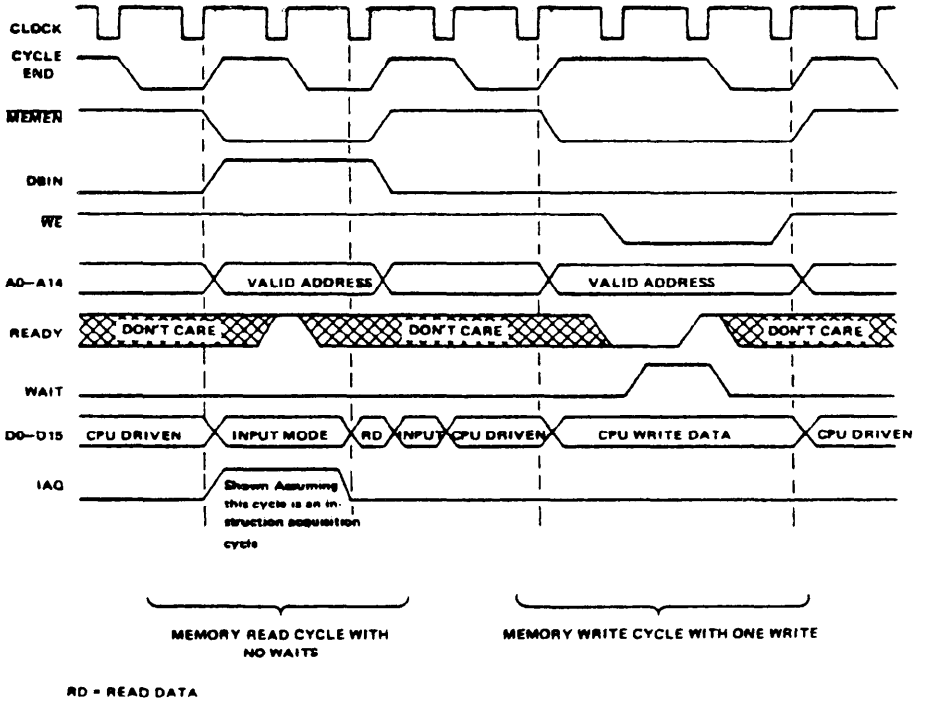


FIGURE 22. Memory bus timing.

In the case of a memory-read cycle, DBIN becomes active (pulled to logic level HIGH) at the same time memory-address data becomes valid; the memory write strobe $\overline{\text{WE}}$ remains inactive (pulled to logic level HIGH). If the memory-read cycle is initiated for acquisition of an instruction, IAQ becomes active (pulled to logic level HIGH) at the same time $\overline{\text{MEMEN}}$ becomes active. At the end of a memory-read cycle, $\overline{\text{MEMEN}}$ and DBIN together become inactive. At that time, though the address may change, the data bus remains in the input mode until terminated by the next high-to-low transition of the clock.

In the case of a memory-write cycle, $\overline{\text{WE}}$ becomes active (logic level LOW) with the first high-to-low transition of the clock after $\overline{\text{MEMEN}}$ becomes active; DBIN remains inactive. At the end of a memory-write cycle, $\overline{\text{WE}}$ and $\overline{\text{MEMEN}}$ together become inactive.

During either a memory-read or a memory-write operation, READY may be used to extend the duration of the associated memory cycle such that the speed of the memory system may be coordinated with the speed of the device. If READY is inactive (logic level LOW) during the first low-to-high transition of the clock after $\overline{\text{MEMEN}}$ becomes active, the device will enter a wait state suspending further progress of the memory cycle. The first low-to-high transition of the clock after READY becomes active terminates the wait state and allows normal completion of the memory cycle.

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40.2 Hold. The hold facilities allow both the microprocessor and external devices to share a common memory. To gain memory-bus control, an external device requiring direct memory access (DMA) sends a hold request ($\overline{HOLD}$) to the device. When the next available non-memory cycle occurs, the device enters a hold state and signals its surrender of the memory-bus to the external device via a hold acknowledge (HOLDA). Receiving the hold acknowledgement, the external device proceeds to utilize the common memory. After its memory requirements have been satisfied, the external device returns memory-bus control to the device by releasing $\overline{HOLD}$.

When $\overline{HOLD}$ becomes active (logic level LOW), the device enters a hold state at the beginning of the next available non-memory cycle as shown in figure 23. Upon entering a hold state, HOLDA becomes active (pulled to logic level HIGH) with the following signals pulled to a HIGH logic level by the individual pull-up resistors tied to each respective open-collector output: DBIN, $\overline{MEMEN}$, $\overline{WE}$, A0 through A14, and D0 through D15. When $\overline{HOLD}$ becomes inactive, the device exits the hold state and regains memory-bus control. If $\overline{HOLD}$ becomes active during a CRU operation, the device uses an extra clock cycle after the deactivation of $\overline{HOLD}$ to reassert the CRU address thereby providing the normal setup time for the CRU-bit transfer.

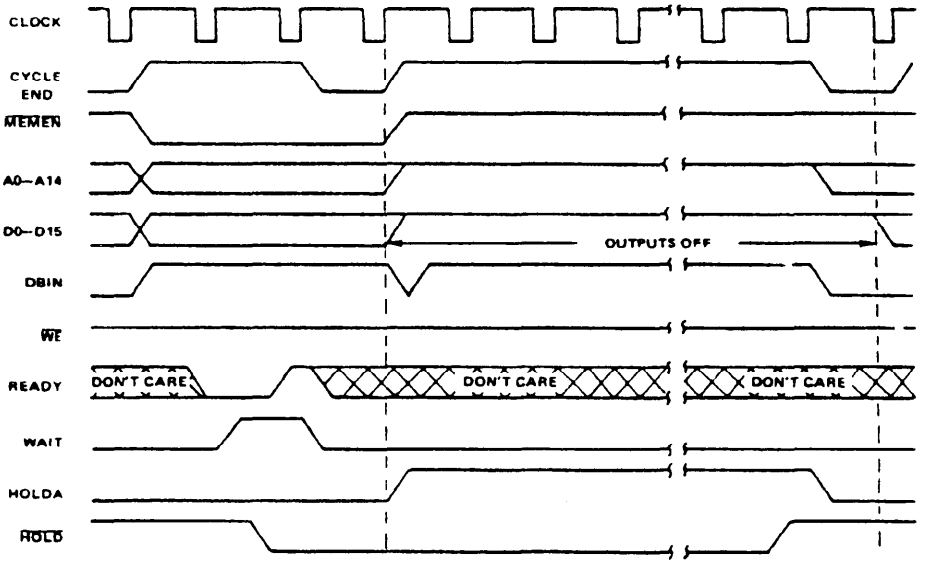


FIGURE 23. Hold timing.

40.3 CRU. The transfer of two data-bits from memory to a peripheral CRU device, and the transfer of one data-bit from a peripheral CRU device to memory, is shown in figure 24. To transfer a data bit to a peripheral CRU device, the device outputs the corresponding CRU-bit-address on address bus bits A3 through A14 and the respective data-bit on CRUOUT. During the second clock cycle of the operation, the device outputs a pulse, on CRUCLK, indicating to the peripheral CRU device the presence of a data-bit. This process is repeated until transfer of the entire field of data-bits specified by the CRU instruction has been accomplished. To transfer a data-bit from a peripheral CRU device, the device outputs the corresponding CRU-bit-Address on address bus bits A3 through A14 and receives the respective data-bit on CRUIN. No CRUCLK pulses occur during a CRU input operation.

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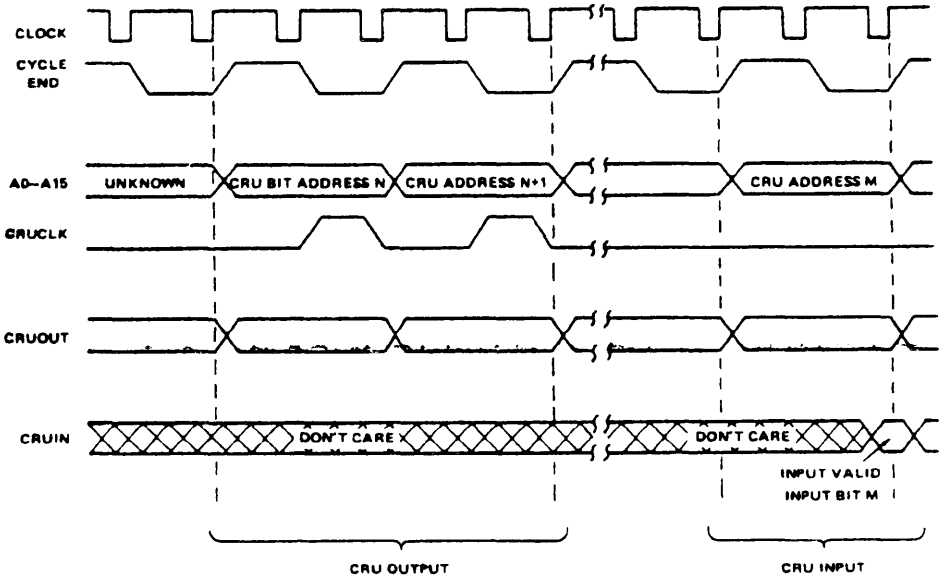
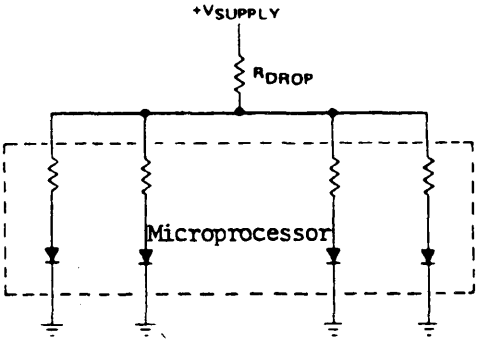


FIGURE 24. CRU interface timing.



GENERAL FORMULA (OHM'S LAW)

$$R_{DROP} = \frac{V_{SUPPLY} - V_{CC}}{I_{CC}}$$

EXAMPLE FOR V_{SUPPLY} = 5V, AND I_{CC} = 500 mA:

$$R_{DROP} = \frac{5 - 1.05}{0.5} = \frac{3.95}{0.5} = 7.9 \text{ OHMS}$$

FIGURE 25. Injector current calculations.

50. POWER SOURCE

50.1 I^2L is a current-injected logic. When placed across a curve tracer, the processor will resemble a silicon switching diode. Any voltage or current source capable of supplying the desired current at the injector node voltage required will suffice (see figures 25 and 26).

The output circuit characteristic for logic-level low output voltage (V_{OL}) versus logic-level low output current (I_{OL}) is shown in figure 27. At rated injector current, the output circuit offers a low-level output voltage of typically 220 mV.

The output circuit characteristics for: (1) logic-level high output voltage (V_{OH}) and current (I_{OH}); (2) rise times; and (3) next stage input noise immunity, are a function of the load circuit being sourced. The load circuit may be either the direct input, if no source current is required, of a 5-volt logic family function, or, for greater noise immunity and improved rise times, the direct input of a 5-volt logic family function in conjunction with a discrete pull-up resistor.

When a discrete pull-up resistor (R_L) is utilized, the fanout requirements placed on a particular output restrict both the maximum and minimum value of R_L.

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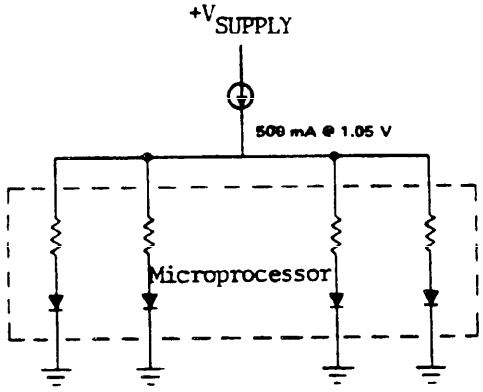


FIGURE 26. Injector current source.

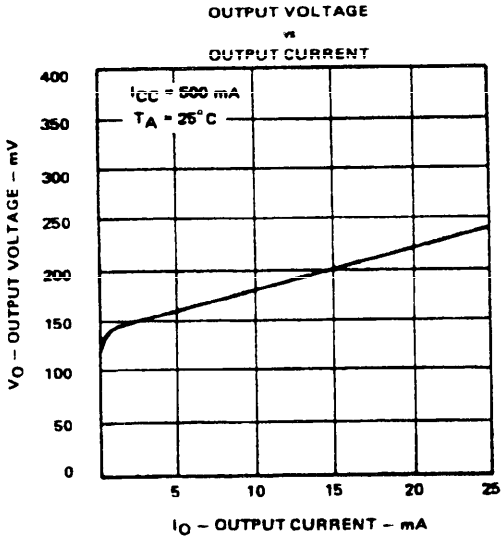


FIGURE 27. Typical output characteristics.

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Operating from a constant current power source, the device may be powered-up/powered-down with complete maintenance of data integrity to execute instructions over a speed/power range spanning several orders of user-selectable injector-supply-current range as illustrated in figure 28.

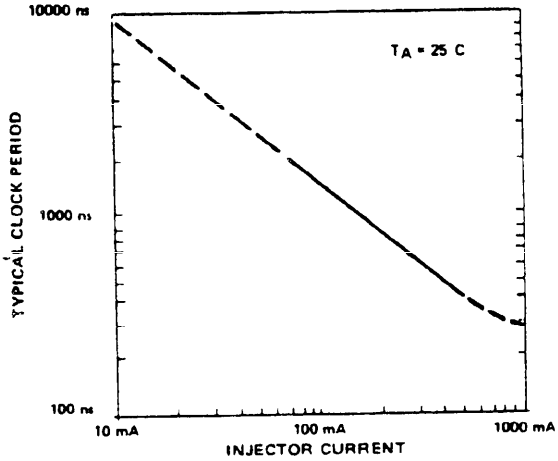


FIGURE 28. Clock period vs injector current.

Figures 29 and 30 show the typical injector node voltages which occur across the temperature and injector current ranges.

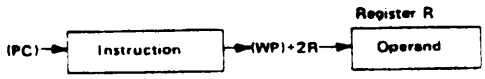
60. INSTRUCTION SET

60.1 Definition. Each device instruction performs one of the following operations:

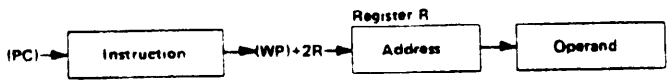
a. Arithmetic, logical, comparison, or manipulation operations on data.
b. Loading or storage of internal registers (program counter, workspace pointer, or status).
c. Data transfer between memory and external devices via the CRU.
d. Control functions.

60.2 Addressing modes. The instructions contain a variety of available modes for addressing random-memory data (e.g., program parameters and flags), or formatted memory data (character strings, data lists, etc.). The following figures graphically describe the derivation of the effective address for each addressing mode. The applicability of addressing modes to particular instructions is described in 60.5 along with the description of the operations performed by the instruction. The symbols following the names of the addressing modes (R, *R, *R+, @LABEL, or @TABLE (R)) are the general forms used by assemblers to select the addressing mode for register R.

60.2.1 Workspace register addressing R. Workspace register R contains the operand.



60.2.2 Workspace register indirect addressing *R. Workspace register R contains the address of the operand.



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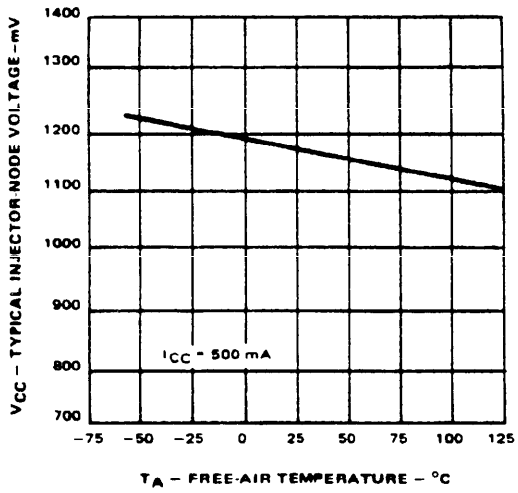


FIGURE 29. Injector-node voltage vs free air temperature.

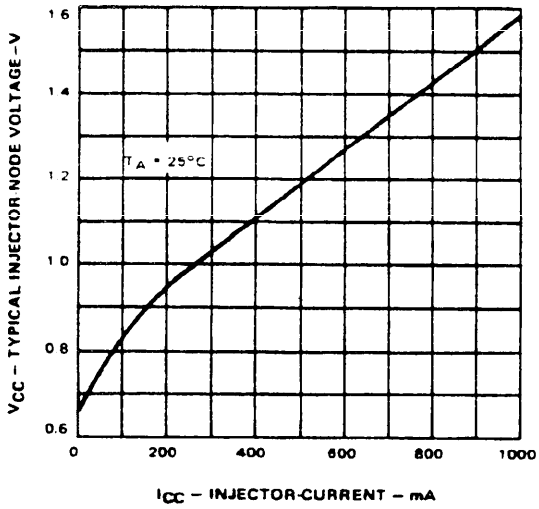
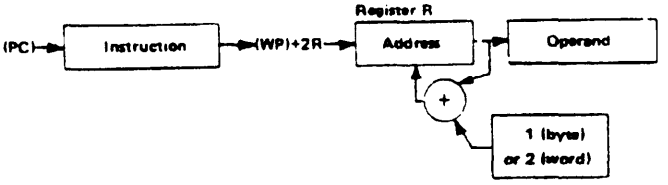


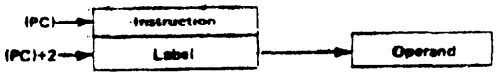
FIGURE 30. Injector-node voltage vs injector current.

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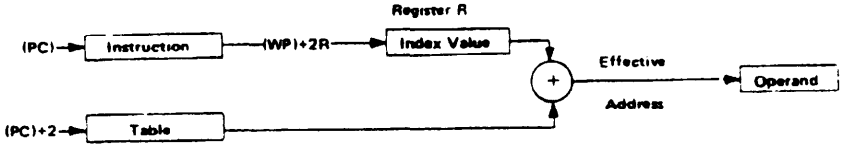
60.2.3 Workspace register indirect auto increment addressing *R+. Workspace register R contains the address of the operand. After acquiring the operand, the contents of workspace register R are incremented.



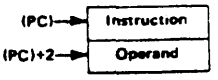
60.2.4 Symbolic (direct) addressing @LABEL. The word following the instruction contains the address of the operand.



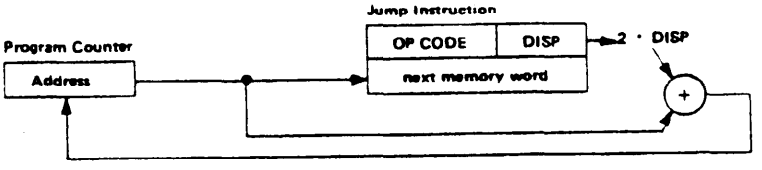
60.2.5 Indexed addressing @TABLE (R). The word following the instruction contains the base address. Workspace register R contains the index value. The sum of the base address and the index value results in the effective address of the operand.



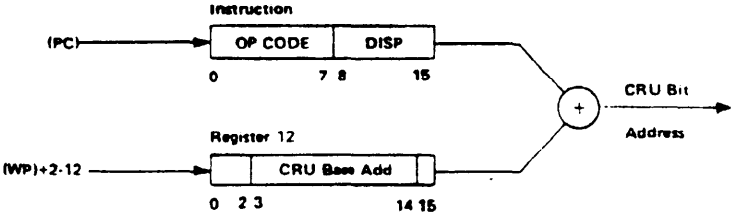
60.2.6 Immediate addressing. The word following the instruction contains the operand.



60.2.7 Program counter relative addressing. The 8-bit signed displacement in the right byte (bits 8 through 15) of the instruction is multiplied by 2 and added to the updated contents of the program counter. The result is placed in the PC.



60.2.8 CRU relative addressing. The 8-bit signed displacement in the right byte of the instruction is added to the CRU base address (bits 3 through 14 of the workspace register 12). The result is the CRU address of the selected CRU bit.



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60.3 Terms and definitions. The terms used in describing the instructions are defined in table VII.

TABLE VII. Term definitions.

TERM	DEFINITION
B	Byte indicator (1=byte, 0 = word)
C	Bit count
D	Destination address register
DA	Destination address
IOP	Immediate operand
LSB(n)	Least significant (right most) bit of (n)
MSB(n)	Most significant (left most) bit of (n)
N	Don't care
PC	Program counter
Result	Result of operation performed by instruction
S	Source address register
SA	Source address
ST	Status register
STn	Bit n of status register
T_D	Destination address modifier
T_S	Source address modifier
W	Workspace register
WPn	Workspace register n
(n)	Contents of n
a → b	a is transferred to b
|n|	Absolute value of n
+	Arithmetic addition
-	Arithmetic subtraction
AND	Logical AND
OR	Logical OR
⊕	Logical exclusive OR
$\bar{n}$	Logical complement of n

60.4 Status register. The status register contains the interrupt mask level and information pertaining to the instruction operation. Table VIII explains the bit indications.

0	1	2	3	4	5	6	7 8 9 10 11	12 13 14 15
ST0	ST1	ST2	ST3	ST4	ST5	ST6	not used (=0)	ST12 ST13 ST14 ST15
L>	A>	=	C	O	P	X		Interrupt Mask

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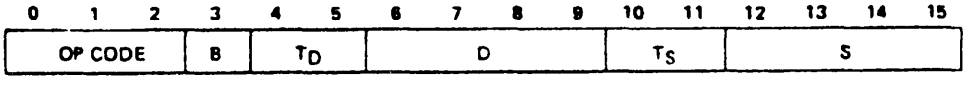
TABLE VIII. Status register bit definitions.

BIT	NAME	INSTRUCTION	CONDITION TO SET BIT TO 1
ST0	LOGICAL GREATER THAN	C,CB	If MSB(SA) = 1 and MSB(DA) = 0, or if MSB(SA) = MSB(DA) and MSB of [(DA)-(SA)] = 1
		CI	If MSB(W) = 1 and MSB of IOP = 0, or if MSB(W) = MSB of IOP and MSB of [IOP-(W)] = 1
		ABS	If (SA) ≠ 0
		All Others	If result ≠ 0
ST1	ARITHMETIC GREATER THAN	C,CB	If MSB(SA) = 0 and MSB(DA) = 1, or if MSB(SA) = MSB(DA) and MSB of [(DA)-(SA)] = 1
		CI	If MSB(W) = 0 and MSB of IOP = 1, or if MSB(W) = MSB of IOP and MSB of [IOP-(W)] = 1
		ABS	If MSB(SA) = 0 and (SA) ≠ 0
		All Others	If MSB of result = 0 and result ≠ 0
ST2	EQUAL	C, CB	If (SA) = (DA)
		C1	If (W) = IOP
		COC	If (SA) and $\overline{(DA)}$ = 0
		CZC	If (SA) and (DA) = 0
		TB	If CRUIN = 1
		ABS	If (SA) = 0
		All Others	If result = 0
ST3	CARRY	A, AB, ABS, AI, DEC, DECT, INC, INCT, NEG, S, SB	If CARRY OUT = 1
		SLA, SRA, SRC, SRL	If last bit shifted out = 1
ST4	OVERFLOW	A, AB	If MSB(SA) = MSB(DA) and MSB of result ≠ MSB(DA)
		AI	If MSB(W) = MSB of IOP and MSB of result ≠ MSB(W)
		S, SB	If MSB(SA) ≠ MSB(DA) and MSB of result ≠ MSB(DA)
		DEC, DECT	If MSB(SA) = 1 and MSB of result = 0
		INC, INCT	If MSB(SA) = 0 and MSB of result = 1
		SLA	If MSB changes during shift
		DIV	If MSB(SA) = 0 and MSB(DA) = 1, or if MSB(SA) = MSB(DA) and MSB of [(DA)-(SA)] = 0
		ABS, NEG	If (SA) = 8000_{16}
ST5	PARITY	CB, MOVB	If (SA) has odd number of 1's
		LDCR, STCR	If 1 < C < 8 and (SA) has odd number of 1's
		AB, SB, SOCB, SZCB	If result has odd number of 1's
ST6	XOP	XOP	If XOP instruction is executed
ST12–ST15	INTERRUPT MASK	LIMI	If corresponding bit of IOP is 1
		RTWP	If corresponding bit of WR15 is 1

60.5 Instructions.

60.5.1 Dual operand instructions with multiple addressing modes for source and destination operand.

General format:



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If B = 1 the operands are bytes and the operand addresses are byte addresses. If B = 0 the operands are words and the operand addresses are word addresses.

The addressing mode for each operand is determined by the T field of that operand.

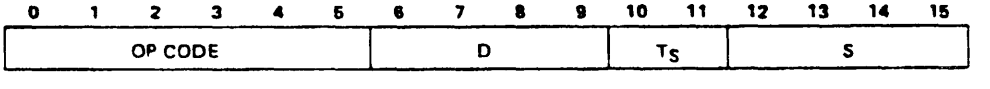
T_S OR T_D	S OR D	ADDRESSING MODE	NOTES
00	0, 1, . . . 15	Workspace register	1
01	0, 1, . . . 15	Workspace register indirect	
10	0	Symbolic	4
10	1, 2, . . . 15	Indexed	2,4
11	0, 1, . . . 15	Workspace register indirect auto-increment	3

NOTES: 1. When a workspace register is the operand of a byte instruction (bit 3 = 1), the left byte (bits 0 through 7) is the operand and the right byte (bits 8 through 15) is unchanged.
2. Workspace register 0 may not be used for indexing.
3. The workspace register is incremented by 1 for byte instructions (bit 3 = 1) and is incremented by 2 for word instructions (bit 3 = 0).
4. When T_S = T_D = 10, two words are required in addition to the instruction word. The first word is the source operand base address and the second word is the destination operand base address.

MNEMONIC	OP CODE 0 1 2	B 3	MEANING	RESULT COMPARED TO 0	STATUS BITS AFFECTED	DESCRIPTION
A	1 0 1	0	Add	Yes	0-4	(SA)+(DA) → (DA)
AB	1 0 1	1	Add bytes	Yes	0-5	(SA)+(DA) → (DA)
C	1 0 0	0	Compare	No	0-2	Compare (SA) to (DA) and set appropriate status bits
CB	1 0 0	1	Compare bytes	No	0-2,5	Compare (SA) to (DA) and set appropriate status bits
S	0 1 1	0	Subtract	Yes	0-4	(DA) – (SA) → (DA)
SB	0 1 1	1	Subtract bytes	Yes	0-5	(DA) – (SA) → (DA)
SOC	1 1 1	0	Set ones corresponding	Yes	0-2	(DA) OR (SA) → (DA)
SOCB	1 1 1	1	Set ones corresponding bytes	Yes	0-2,5	(DA) OR (SA) → (DA)
SZC	0 1 0	0	Set zeroes corresponding	Yes	0-2	(DA) AND ($\overline{SA}$) → (DA)
SZCB	0 1 0	1	Set zeroes corresponding bytes	Yes	0-2,5	(DA) AND ($\overline{SA}$) → (DA)
MOV	1 1 0	0	Move	Yes	0-2	(SA) → (DA)
MOVB	1 1 0	1	Move bytes	Yes	0-2,5	(SA) → (DA)

60.5.2 Dual operand instructions with multiple addressing modes for the source operand and workspace register addressing for the destination.

General format:



The addressing mode for the source operand is determined by the T_S field.

T_S	S	ADDRESSING MODE	NOTES
00	0, 1, . . . 15	Workspace register	
01	0, 1, . . . 15	Workspace register indirect	
10	0	Symbolic	
10	1, 2, . . . 15	Indexed	1
11	0, 1, . . . 15	Workspace register indirect auto increment	2

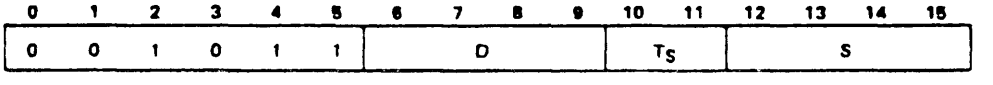
NOTES: 1. Workspace register 0 may not be used for indexing.
2. The workspace register is incremented by 2.

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MNEMONIC	OP CODE 0 1 2 3 4 5	MEANING	RESULT COMPARED TO 0	STATUS BITS AFFECTED	DESCRIPTION
COC	0 0 1 0 0 0	Compare ones corresponding	No	2	Test (D) to determine if 1's are in each bit position where 1's are in (SA). If so, set ST2.
CZC	0 0 1 0 0 1	Compare zeros corresponding	No	2	Test (D) to determine if 0's are in each bit position where 1's are in (SA). If so, set ST2.
XOR	0 0 1 0 1 0	Exclusive OR	Yes	0-2	(D) ⊕ (SA) → (D)
MPY	0 0 1 1 1 0	Multiply	No		Multiply unsigned (D) by unsigned (SA) and place unsigned 32-bit product in D (most significant) and D+1 (least significant). If WR15 is D, the next word in memory after WR15 will be used for the least significant half of the product.
DIV	0 0 1 1 1 1	Divide	No	4	If unsigned (SA) is less than or equal to unsigned (D), perform no operation and set ST4. Otherwise, divide unsigned (D) and (D+1) by unsigned (SA). Quotient → (D), remainder → (D+1). If D = 15, the next word in memory after WR 15 will be used for the remainder.

60.5.3 Extended operation (XOP) instruction.

General format:



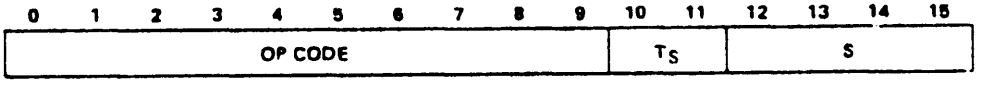
The T_S and S fields provide multiple mode addressing capability for the source operand. When the XOP is executed, ST8 is set and the following transfers occur:

$(40_{16} + 4D) \rightarrow$ (WP)
$(42_{16} + 4D) \rightarrow$ (PC)
SA → (new WR11)
(old WP) → (new WR13)
(old PC) → (new WR14)
(old ST) → (new WR15)

The device does not test interrupt requests ($\overline{\text{INTREQ}}$) upon completion of the XOP instruction.

60.5.4 Single operand instructions.

General format:



The T_S and S fields provide multiple mode addressing capability for the source operand.

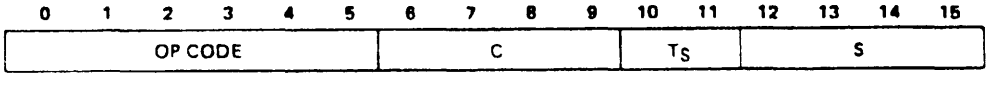
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MNEMONIC	OP CODE 0 1 2 3 4 5 6 7 8 9	MEANING	RESULT COMPARED TO 0	STATUS BITS AFFECTED	DESCRIPTION
B	0 0 0 0 0 1 0 0 0 1	Branch	No	–	SA → (PC)
BL	0 0 0 0 0 1 1 0 1 0	Branch and link	No		(PC) → (WR11), SA → (PC)
BLWP	0 0 0 0 0 1 0 0 0 0	Branch and load workspace pointer	No	–	(SA) → (WP); (SA+2) → (PC); (old WP) → (new WR13); (old PC) → (new WR14); (old ST) → (new WR15); the interrupt input ($\overline{\text{INTREQ}}$) is not tested upon completion of the BLWP instruction.
CLR	0 0 0 0 0 1 0 0 1 1	Clear operand	No	–	0 → (SA)
SETO	0 0 0 0 0 1 1 1 0 0	Set to ones	No	–	$FFFF_{16}$ → (SA)
INV	0 0 0 0 0 1 0 1 0 1	Invert	Yes	0-2	$(\overline{SA})$ → (SA)
NEG	0 0 0 0 0 1 0 1 0 0	Negate	Yes	0-4	−(SA) → (SA)
ABS	0 0 0 0 0 1 1 1 0 1	Absolute value*	No	0-4	|(SA)| → (SA)
SWPB	0 0 0 0 0 1 1 0 1 1	Swap bytes	No	–	(SA), bits 0 thru 7 → (SA), bits 8 thru 15; (SA), bits 8 thru 15 → (SA), bits 0 thru 7
INC	0 0 0 0 0 1 0 1 1 0	Increment	Yes	0-4	(SA) + 1 → (SA)
INCT	0 0 0 0 0 1 0 1 1 1	Increment by two	Yes	0-4	(SA) + 2 → (SA)
DEC	0 0 0 0 0 1 1 0 0 0	Decrement	Yes	0-4	(SA) − 1 → (SA)
DECT	0 0 0 0 0 1 1 0 0 1	Decrement by two	Yes	0-4	(SA) − 2 → (SA)
X†	0 0 0 0 0 1 0 0 1 0	Execute	No	–	Execute the instruction at SA.

*Operand is compared to zero for status bit.
†If additional memory words for the execute instruction are required to define the operands of the instruction located at SA, these words will be accessed from PC and the PC will be updated accordingly. The instruction acquisition signal (IAQ) will not be true when the device accesses the instruction at SA. Status bits are affected in the normal manner for the instruction executed.

60.5.5 CRU multiple-bit instructions.

General format:



The C field specifies the number of bits to be transferred. If C = 0, 16 bits will be transferred. The CRU base register (WR12, bits 3 through 14) defines the starting CRU bit address. The bits are transferred serially and the CRU address is incremented with each bit transfer, although the contents of WR12 is not affected. T_S and S provide multiple mode addressing capability for the source operand. If 8 or fewer bits are transferred (C = 1 through 8), the source address is a byte address. If 9 or more bits are transferred (C = 0, 9 through 15), the source address is a word address. If the source is addressed in the workspace register indirect auto increment mode, the workspace register is incremented by 1 if C = 1 through 8, and is incremented by 2 otherwise.

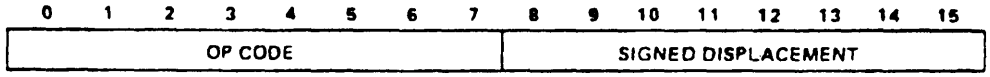
MNEMONIC	OP CODE 0 1 2 3 4 5	MEANING	RESULT COMPARED TO 0	STATUS BITS AFFECTED	DESCRIPTION
LDCR	0 0 1 1 0 0	Load communcation register	Yes	0-2,5†	Beginning with LSB of (SA), transfer the specified number of bits from (SA) to the CRU.
STCR	0 0 1 1 0 1	Store communcation register	Yes	0-2,5†	Beginning with LSB of (SA), transfer the specified number of bits from the CRU to (SA). Load unfilled bit positions with 0.

†ST5 is affected only if 1 ≤ C ≤ 8.

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60.5.6 CRU single-bit instructions.

General format:

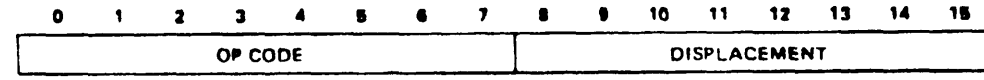


CRU relative addressing is used to address the selected CRU bit.

MNEMONIC	OP CODE 0 1 2 3 4 5 6 7	MEANING	STATUS BITS AFFECTED	DESCRIPTION
SBO	0 0 0 1 1 1 0 1	Set bit to one	–	Set the selected CRU output bit to 1.
SBZ	0 0 0 1 1 1 1 0	Set bit to zero	–	Set the selected CRU output bit to 0.
TB	0 0 0 1 1 1 1 1	Test bit	2	If the selected CRU input bit = 1, set ST2.

60.5.7 Jump instructions.

General format:



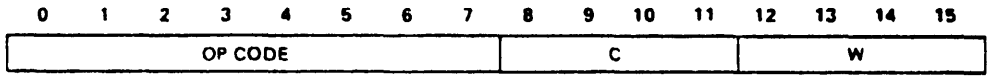
Jump instructions cause the PC to be loaded with the value selected by PC relative addressing if the bits of ST are at specified values. Otherwise, no operation occurs and the next instruction is executed since PC points to the next instruction. The displacement field is a word count to be added to PC. Thus, the jump instruction has a range of –128 to 127 words from memory-word address following the jump instruction. No ST bits are affected by jump instruction.

MNEMONIC	OP CODE 0 1 2 3 4 5 6 7	MEANING	ST CONDITION TO LOAD PC
JEQ	0 0 0 1 0 0 1 1	Jump equal	ST2 = 1
JGT	0 0 0 1 0 1 0 1	Jump greater than	ST1 = 1
JH	0 0 0 1 1 0 1 1	Jump high	ST0 = 1 and ST2 = 0
JHE	0 0 0 1 0 1 0 0	Jump high or equal	ST0 = 1 or ST2 = 1
JL	0 0 0 1 1 0 1 0	Jump low	ST0 = 0 and ST2 = 0
JLE	0 0 0 1 0 0 1 0	Jump low or equal	ST0 = 0 or ST2 = 1
JLT	0 0 0 1 0 0 0 1	Jump less than	ST1 = 0 and ST2 = 0
JMP	0 0 0 1 0 0 0 0	Jump unconditional	unconditional
JNC	0 0 0 1 0 1 1 1	Jump no carry	ST3 = 0
JNE	0 0 0 1 0 1 1 0	Jump not equal	ST2 = 0
JNO	0 0 0 1 1 0 0 1	Jump no overflow	ST4 = 0
JOC	0 0 0 1 1 0 0 0	Jump on carry	ST3 = 1
JOP	0 0 0 1 1 1 0 0	Jump odd parity	ST5 = 1

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60.5.8 Shift instructions.

General format:

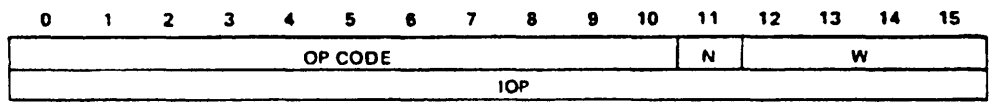


If C = 0, bits 12 through 15 of WR0 contain the shift count. If C = 0 and bits 12 through 15 of WR0 = 0, the shift count is 16.

MNEMONIC	OP CODE 0 1 2 3 4 5 6 7	MEANING	RESULT COMPARED TO 0	STATUS BITS AFFECTED	DESCRIPTION
SLA	0 0 0 0 1 0 1 0	Shift left arithmetic	Yes	0-4	Shift (W) left. Fill vacated bit positions with 0.
SRA	0 0 0 0 1 0 0 0	Shift right arithmetic	Yes	0-3	Shift (W) right. Fill vacated bit positions with original MSB of (W).
SRC	0 0 0 0 1 0 1 1	Shift right circular	Yes	0-3	Shift (W) right. Shift previous LSB into MSB.
SRL	0 0 0 0 1 0 0 1	Shift right logical	Yes	0-3	Shift (W) right. Fill vacated bit positions with 0's.

60.5.9 Immediate register instructions.

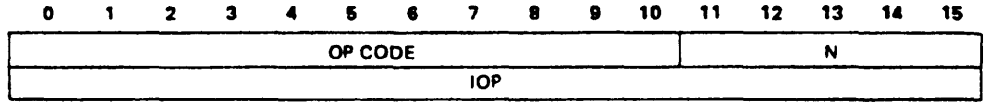
General format:



MNEMONIC	OP CODE 0 1 2 3 4 5 6 7 8 9 10	MEANING	RESULT COMPARED TO 0	STATUS BITS AFFECTED	DESCRIPTION
AI	0 0 0 0 0 0 1 0 0 0 1	Add immediate	Yes	0-4	(W) + IOP → (W)
ANDI	0 0 0 0 0 0 1 0 0 1 0	AND immediate	Yes	0-2	(W) AND IOP → (W)
CI	0 0 0 0 0 0 1 0 1 0 0	Compare immediate	Yes	0-2	Compare (W) to IOP and set appropriate status bits
LI	0 0 0 0 0 0 1 0 0 0 0	Load immediate	Yes	0-2	IOP → (W)
ORI	0 0 0 0 0 0 1 0 0 1 1	OR immediate	Yes	0-2	(W) OR IOP → (W)

60.5.10 Internal register load immediate instructions.

General format:

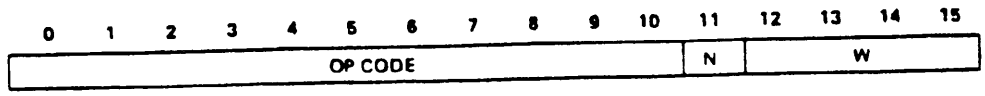


MNEMONIC	OP CODE 0 1 2 3 4 5 6 7 8 9 10	MEANING	DESCRIPTION
LWPI	0 0 0 0 0 0 1 0 1 1 1	Load workspace pointer immediate	IOP → (WP), no ST bits affected
LIMI	0 0 0 0 0 0 1 1 0 0 0	Load interrupt mask	IOP, bits 12 thru 15 → ST12 thru ST15

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60.5.11 Internal register store instructions.

General format:



No ST bits are affected.

MNEMONIC	OP CODE 0 1 2 3 4 5 6 7 8 9 10	MEANING	DESCRIPTION
STST	0 0 0 0 0 0 1 0 1 1 0	Store status register	(ST) → (W)
STWP	0 0 0 0 0 0 1 0 1 0 1	Store workspace pointer	(WP) → (W)

60.5.12 Return workspace pointer (RTWP) instruction.

General format:

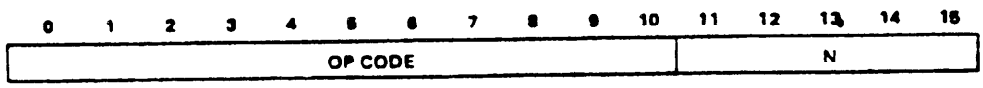
0	1	2	3	4	5	6	7	8	9	10	11 12 13 14 15
0	0	0	0	0	0	1	1	1	0	0	N

The RTWP instruction causes the following transfers to occur:

(WR15) → (ST)
(WR14) → (PC)
(WR13) → (WP)

60.5.13 External instructions.

General format:



External instructions cause the three most-significant address lines (A0 through A2) to be set to the below-described levels and the CRUCLK line to be pulsed, allowing external control functions to be initiated.

MNEMONIC	OP CODE 0 1 2 3 4 5 6 7 8 9 10	MEANING	STATUS BITS AFFECTED	DESCRIPTION	ADDRESS BUS A0	A1	A2
IDLE	0 0 0 0 0 0 1 1 0 1 0	Idle	–	Suspend TMS 9900 instruction execution until an interrupt, $\overline{\text{LOAD}}$, or $\overline{\text{RESET}}$ occurs	L	H	L
RSET	0 0 0 0 0 0 1 1 0 1 1	Reset	12–15	0 → ST12 thru ST15	L	H	H
CKOF	0 0 0 0 0 0 1 1 1 1 0	User defined		---	H	H	L
CKON	0 0 0 0 0 0 1 1 1 0 1	User defined		---	H	L	H
LREX	0 0 0 0 0 0 1 1 1 1 1	User defined		---	H	H	H

60.6 Microinstruction cycle. The device includes circuitry which will indicate the completion of a microinstruction cycle. Designated as the $\overline{\text{CYCEND}}$ function, it provides CPU status that can simplify system design. The $\overline{\text{CYCEND}}$ output will go to a low logic level as a result of the low-to-high transition of each clock pulse which initiates the last clock of a microinstruction.

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TABLE IX. Instruction execution times.

Instruction	Clock cycles C	Memory access M	Address modification+ Source	Address modification+ Dest
A	14	4	A	A
AB	14	4	A	A
ABS (MSB = 0)	12	2	A	-
(MSB = 1)	14	3	A	-
AI	14	4	-	-
ANDI	14	4	-	-
B	8	2	A	-
BL	12	3	A	-
BLWP	26	6	A	-
C	14	3	A	A
CB	14	3	A	A
CI	14	3	-	-
CKOF	12	1	-	-
CKON	12	1	-	-
CLR	10	3	A	-
COC	14	3	A	-
CZC	14	3	A	-
DEC	10	3	A	-
DECT	10	3	A	-
DIV (ST4 is set)	16	3	A	-
DIV (ST4 is reset)*	92-124	6	A	-
IDLE	12	1	-	-
INC	10	3	A	-
INCT	10	3	A	-
INV	10	3	A	-
Jump (PC is changed)	10	1	-	-
(PC is not changed	8	1	-	-
LDCR (C=0)	52	3	A	-
(1<C<8)	20+2C	3	A	-
(9<C<15)	20+2C	3	A	-
LI	12	3	-	-
LIMI	14	2	-	-
LREX	12	1	-	-
$\overline{\text{RESET}}$ function	26	5	-	-
$\overline{\text{LOAD}}$ function	22	5	-	-
Interrupt context switch	22	5	-	-
LWPI	10	2	-	-
MOV	14	4	A	A
MOVB	14	4	A	A
MPY	52	5	A	-
NEG	12	3	A	-
ORI	14	4	-	-
RSET	12	1	-	-
RTWP	14	4	-	-
S	14	4	A	A
SB	14	4	A	A
SBO	12	2	-	-
SBZ	12	2	-	-
SETO	10	3	A	-
Shift (C=0)	12+2C	3	-	-
(C=0, Bits 12-15 of WR0=0)	52	4	-	-
(C=0, Bits 12-15 of WRP=N=0)	20+2N	4	-	-
SOC	14	4	A	A
SOCB	14	4	A	A
STCR (C=0)	60	4	A	-
(1<C<7)	42	4	A	-
(C=8)	44	4	A	-
(9<C<15)	58	4	A	-
STST	8	2	-	-
STWP	8	2	-	-
SWPB	10	3	A	-
SZC	14	4	A	A
SZCB	14	4	A	A
TB	12	2	-	-
X **	4	1	A	-
XOP	36	8	A	-
XOR	14	4	A	-
Undefined op codes 0000-01FF, 0320-033F, 0C00-0FFF, 0780-07FF	6	1	-	-

*Execution time is dependent upon the partial quotient after each clock cycle during execution.
**Execution time is added to the execution time of the instruction located at the source address minus 4 clock cycles and 1 memory access time.
+The letter A refers to table IXA.

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60.7 Instruction execution times. Instruction execution times for the device are a function of:

a. Clock cycle time, t_c.
b. Addressing mode used where operands have multiple addressing mode capability.
c. Number of wait states required per memory access.

Table IX lists the number of clock cycles and memory accesses required to execute each instruction. For instructions with multiple addressing modes for either or both operands, the table lists the number of clock cycles and memory accesses with all operands addressed in the workspace-register mode. To determine the additional number of clock cycles and memory accesses required for modified addressing, add the appropriate values from table IXA. The total instruction-execution time for an instruction is:

$$T = t_c (C + W \cdot M)$$

where:
T = total instruction execution time:
t_c = clock cycle time;
C = number of clock cycles for instruction execution plus address modification;
W = number of required wait states per memory access for instruction execution plus address modification;
M = number of memory accesses.

TABLE IXA. Address modification.

Addressing mode	Clock cycles C	Memory accesses M
WR (T_S or T_D = 00)	0	0
WR indirect (T_S or T_D = 01)	4	1
WR indirect auto-increment (T_S or T_D = 11)		
Word operation	8	2
Byte operation	6	2
Symbolic (T_S or T_D = 10, S or D = 0)	8	1
Indexed (T_S or T_D = 10, S or D ≠ 0)	8	2

As an example, the instruction MOVB is used in a system with t_c = 0.5 µs and no wait states are required to access memory. Both operands are addressed in the workspace register mode:

$$T = t_c (C + W \cdot M) = 0.5 (14 + 0 \cdot 4)\ \mu s = 7\ \mu s.$$

If two wait states per memory access were required, the execution time is:

$$T = 0.5 (14 + 2 \cdot 4)\ \mu s = 11\ \mu s.$$

If the source operand was addressed in the symbolic mode and two wait states were required:

$$T = t_c (C + W \cdot M)$$
$$C = 14 + 8 = 22$$
$$M = 4 + 1 = 5$$
$$T = 0.5 (22 + 2 \cdot 5)\ \mu s = 16\ \mu s$$

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